

Benchmarking the GT Screener Against CogAT

A staged predictive-validity design, its power, and its limits

GT Selection Capstone — methodology track

July 31, 2026

Abstract

GT is building a cognitive screener and needs to know whether it beats CogAT at the only thing that matters: identifying children who go on to thrive in a personalised-mastery programme. This report specifies how to answer that question and, more importantly, reports which parts of the answer are within reach at the programme's current size and which are not. The criterion is fixed as mastery pace, anchored to NWEA MAP conditional growth percentiles. The study is split into a stage that can run now and a stage that waits on CogAT data. Companion simulation work has since measured the effect sizes involved, and the result reshapes the recommendation: **a head-to-head superiority claim against CogAT is not executable at the current scale** — it needs roughly 520 to 2590 children against an effective sample of 157 — while the better-posed question of whether the screener can *replace* CogAT is close to reachable now and clearly reachable across two or three admission cycles. The choice of estimand turns out to matter more than the sample: framing the comparison as an AUC difference rather than a variance increment multiplies the required sample by about 3.7. Two assumptions carry the argument and neither has been measured — the correlation between standing ability and learning rate, and the comparator's reliability — and both are flagged as contestable rather than settled. Every number here is generated by a self-testing script in `scripts/`; none is asserted.

Claim boundary. This is a *predictive-validity* design. It can establish whether a screener score forecasts how a child does in the programme. It cannot establish that the programme caused that outcome, and no analysis in this document should ever be described as measuring GT's impact. Section 2 states the line precisely.

Simulated data. Every effect size quoted in this report — the AUC values, the variance increments, the routing drift, the officer-agreement figures — comes from **simulated** data produced by a companion synthetic-cohort analysis. They are design inputs used to size a real study. They are not findings about real children and must never be quoted as evidence that the screener works.

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Part I — For decision-makers

1 Executive summary

1.1 What GT can do now

Start the validity study this school year, against mastery pace and MAP growth, on the children already enrolled. Do not wait for CogAT. This is executable with data GT already collects, it is adequately powered on the existing 346 enrolled children, and it produces a defensible statement about the screener on its own. It is also time-limited: the open admission policy that makes it clean will close if GT starts rationing seats (Section 8).

Three things are executable immediately:

1. **Stage 1 — does the screener predict who thrives?** On the pooled cohort, a true validity of 0.35 is estimated to within about ± 0.10 ($[0.25, 0.44]$). Section 4.
2. **The contamination check.** Whether mastery pace is corrupted by the platform’s own routing is testable now, at power 0.87 on the existing cohort. This matters more than it sounds: Section 3.4.
3. **The officer-agreement analysis,** the day historical decisions are released. The record format it needs is specified in Section 4.5.

1.2 What GT cannot do now, and what it would take

The head-to-head claim “our screener beats CogAT” is not reachable at the programme’s current size, and will not be for two to three years. The advantage to be detected is about 0.034 of AUC. Detecting it requires 520–2590 children depending on how similar the two instruments are. The effective sample available today, after accounting for children clustered within campuses and guides, is 157. This is the most important practical conclusion in the report.

At the 46-child campus cohort the confidence interval on a single AUC has a half-width of 0.169, against an effect of 0.034 — the interval is 5.0 times the effect. The study cannot see its own effect.

But 46 is not a fixed ceiling, and the interesting question is what the addressable sample actually is. Section 5 works through every route — one campus, all campuses after the planned expansion, the virtual cohort, multiple admission cycles pooled, and the applicant population at target scale — and prices each in comparability. The short answer is that no single-cycle route reaches the superiority requirement, three pooled cycles reach the variance-increment requirement but not the AUC one, and only the applicant population at target scale reaches both.

Two changes make the comparison affordable, and neither involves recruiting more children. First, ask whether the screener can *replace* CogAT rather than whether it beats it: a non-inferiority test at a 0.02 AUC margin needs 207 children rather than 520. Second, analyse the continuous criterion instead of dichotomising it into “excels / does not”: the variance-increment test needs 222 children against the artefact floor, roughly 3.7 fewer than the AUC comparison. The estimand is a bigger lever than the sample size, and it is free.

1.3 What the simulation found, read honestly

Simulated data. Simulated cohort, not real children.

The screener reaches an AUC of 0.670 [0.651, 0.688] against mastery pace; the CogAT-like comparator reaches 0.636 [0.617, 0.656]; the paired difference is +0.034 [+0.020, +0.049]. Both instruments together reach only 0.672.

The honest reading is “not worse, marginally better.” An AUC of 0.670 is a modest instrument. The advantage over the comparator is real in the simulation but small, and combining the two instruments adds almost nothing to either. This should not be written up as a win, and any GT-facing material that describes the screener as outperforming CogAT is ahead of the evidence — which at present is simulated evidence in any case.

There is, however, a genuine asymmetry worth naming. Adding the screener to the comparator raises explained variance by 0.0495; adding the comparator to the screener raises it by 0.0030, which is below the floor that measurement error alone would produce. In the simulation the screener largely subsumes the comparator, not the other way round. That is the strongest defensible form of GT’s claim, and it is a *replacement* argument, not a superiority one.

1.4 Two assumptions that could overturn this

A hostile reader will find these, so they are stated here rather than in a footnote.

1. **The correlation between standing ability and learning rate is assumed to be 0.30, and has never been measured by anyone** — not by this project, not in the literature. The entire case for the two-phase instrument depends on it. At the administered block length the learning-rate readout clears the subscore-reporting bar only while that correlation stays below 0.35; the working assumption sits just 0.05 below the threshold. At 0.50 it already fails, and at 0.90 the argument collapses at every block length tested. Section 9.
2. **The comparator’s reliability is assumed to be 0.94 because this project tried to verify a published figure and could not.** The primary technical manuals were not text-extractable, so the value was swept rather than asserted. It matters: the artefact floor that a fake increment would clear is 0.0043 at a reliability of 0.94 but 0.0188 at 0.90 — a fourfold range driven entirely by an unverified number. Section 6.4.

1.5 The rest, in brief

1. **The criterion is mastery pace, anchored to MAP — and the anchor is not redundant.** In simulation, mastery-pace validity drifted +0.098 AUC purely from changing

how the platform routed children, while MAP conditional growth moved 0.000. That drift is 2.9 times the entire effect the study is trying to detect. A criterion that can move by three times the effect size for reasons unrelated to the child is not a criterion you rely on alone. Section 3.

2. **Accuracy is not the binding constraint — base rate is.** At a 10% base rate, a screener with validity 0.40 selecting the top 10% is right about 27% of the time. That is arithmetic, and it applies to CogAT equally. Section 7.
3. **Not rationing seats is a research asset worth protecting.** A conventional selective programme sees a validity of 0.40 attenuated to 0.18 and needs 250 children instead of 47. Section 8.
4. **Agreeing with the admissions officer is not the goal, and the data says so.** A rule using the screener and MAP matches 86% of her decisions ($\kappa = 0.43$) — but a rule fitted to mimic her, using her own inputs, ceilings at $\kappa = 0.79$. No instrument can do better than that, because the remainder is her own inconsistency. On the files where the rule and the officer disagree, the rule’s picks meet the criterion 40% of the time against her 33%. Section 4.5.

1.6 Traceability

Every claim above is checkable. A reader who wants to attack this document should start at the appendix named beside the claim.

Summary claim	Argument	Derivation
Superiority against CogAT is not reachable now	§5	App. I
Non-inferiority and ΔR^2 framings are cheaper	§5	App. I
Addressable sample by route	§5	App. B
Mastery pace needs an external anchor	§3	App. C
Routing contamination is bounded and testable	§3.4	App. C
Stage 1 is powered at 346; not at 46	§4	App. B
CogAT is a concordance question, not equating	§6	App. D
Unreliability manufactures an increment	§6.4	App. E
Base rate, not accuracy, binds the decision	§7	App. F
Open admission is worth 5.3× the sample	§8	App. G
Learning rate is not reportable at 30 trials	§9	App. H

1.7 What GT must decide, and what GT must supply

Three decisions are required before the study can be specified further, and none is a methodological question:

1. **Superiority or replacement?** Whether GT needs to show the screener *beats* CogAT or only that it can *replace* it. This is the single largest driver of cost and timeline (Section 5).
2. **What counts as “excelling”** — a threshold or a rank. This sets the base rate, and the base rate determines whether the decision arithmetic in Section 7 is favourable or hostile.
3. **What the score is for** — admission, placement, or flagging for review. A score used to *exclude* must clear a far higher bar than one used to *surface* children for a human look.

The data required is listed in Section 10. Most of it is believed to exist already; where that belief is unverified it is marked as an assumption rather than stated as fact.

2 What this design can and cannot support

This section exists because the most likely way for this work to fail is not a statistical error. It is a sentence in a deck that overstates what was found.

The design can support:

- that screener scores predict mastery pace and MAP growth, with a stated confidence interval;
- that the screener adds (or fails to add) predictive information beyond CogAT, once CogAT scores exist and once the artefact floor is accounted for — subject to the sample being reachable, which Section 5 concludes it currently is not for the superiority version of that claim;
- that decisions made at a given cut have stated sensitivity, specificity and positive predictive value at a stated base rate;
- that classification is or is not stable across administrations;
- that predictive accuracy does or does not differ across demographic groups.

The design cannot support:

- that GT *causes* the outcomes observed. Every child in the sample is enrolled; there is no comparison condition. Predictive validity is a statement about ordering children, not about the effect of the programme on any of them.
- that a child who scores highly *would have* done worse elsewhere. That is a counterfactual, and nothing in this design identifies one.
- that growth from autumn to spring is attributable to the programme. Children grow. Conditional growth percentiles control for starting level, not for maturation, schooling in general, or selection into GT.
- that a validity coefficient justifies a cut score. Validity is necessary and not sufficient; the sufficiency argument is the decision arithmetic in Section 7, and it depends on the base rate and on the consequences of each error type.
- that the screener *outperforms* CogAT. At the sample sizes available this is not a claim the study will be able to make in either direction, and an inconclusive comparison is not evidence of equivalence unless it was designed as an equivalence test in advance (Section 5).

Claim boundary. Predictive validity is not programme impact. Before-and-after growth is not a counterfactual. If GT wants an impact claim it needs a different study with a comparison condition — a randomised offer, a waitlist, or a discontinuity — and none of those is available while the programme admits everyone. This is a genuine tension: the open-admission policy that makes the *validity* study unusually clean is exactly what makes an *impact* study impossible.

Part II — The design

3 The criterion problem

Everything downstream depends on this section. A validity coefficient is a statement about the relationship between two things, and if the second thing is badly chosen, the coefficient is uninterpretable no matter how large the sample.

3.1 The ratified choice

The owner has selected **mastery pace** — objectives or lessons mastered per unit time in the personalised-mastery system — as the primary criterion, with **MAP conditional growth percentile**, autumn to spring, as an external anchor. This report treats that as settled and spends its effort on what the choice costs and how to defend it.

3.2 Why the alternatives were not chosen

Candidate	Cost to collect	Likely reliability	Contamination	Gameable?
Mastery pace (<i>primary</i>)	Near zero if logged	High — many events per child	Serious: partly a property of platform routing, not the child	Yes — by the platform’s own tuning, not usually by the child
MAP conditional growth percentile (<i>anchor</i>)	Near zero — already administered 3×/yr	High; MAP overall reliability ≈ 0.95	Low — external to routing; conditions on starting RIT	Low — third-party normed and proctored
Raw MAP RIT gain	Near zero	Lower than CGP — difference scores	High: high starters gain fewer points, so it penalises exactly the children GT selects	Low
Growth on internal assessments	Low	Unknown	High — same system that routes the child also assesses them	Yes
Objectives completed (no time denominator)	Near zero	High	Confounded with enrolment length and attendance	Yes
Time-to-mastery per objective	Near zero	Moderate — heavy-tailed, needs censoring care	Same routing problem as pace	Yes
Persistence / retention	Near zero	Low as a per-child measure — rare, binary, slow	Dominated by family circumstance, fees, relocation	No, but uninformative
Teacher or guide rating	Moderate — staff time	Low–moderate; rater effects usually large	Fatal if raters know the screener score	Yes
Parent rating	Moderate	Low	Reflects satisfaction more than attainment	Yes

Two entries deserve emphasis because they are the traps.

Raw RIT gain is the wrong metric and would understate GT’s case. NWEA’s own documentation notes that high-starting students “generally do not raise their score as much,” and that expected growth compresses in the upper grades. A gifted cohort scored on raw gain will look mediocre for reasons that have nothing to do with the programme or the screener. Conditional growth percentiles exist precisely to solve this, by comparing each child to peers with the same starting RIT.

Teacher rating is the criterion most likely to produce a spuriously excellent result. If a guide knows a child was flagged as gifted, the rating partly measures the flag. Any rating-based criterion must be collected blind to the screener score, and if that cannot be guaranteed the criterion should be dropped rather than caveated.

3.3 Why the anchor is necessary rather than redundant

If mastery pace and MAP growth were near-perfectly correlated, running both would be waste. They are not, and the anchor earns its place three times over.

(a) Conditional growth already solves the starting-level problem. A validity study relates a screener score — which measures standing ability — to a growth outcome. If the growth outcome is itself depressed for high starters, the study builds in a negative bias against precisely the children the screener identifies, and the resulting validity estimate is too low for a reason that has nothing to do with the screener. Regression to the mean produces the same artefact in the opposite direction for low starters. NWEA’s conditional growth percentile is constructed by conditioning on starting RIT, which removes this confound by design rather than by statistical adjustment after the fact. Doing the adjustment ourselves would require us to estimate the growth-on-starting-level relationship from GT’s own small sample; NWEA has estimated it on a normative sample of millions.

(b) It survives the “grading your own homework” objection. This is the audience argument. Mastery pace is a number GT’s own platform computes, using GT’s own definition of an objective, for children GT admitted. Every one of those is a legitimate target for a hostile reviewer, and none of the rebuttals is fully satisfying. MAP is administered by a third party against published national norms. A validity coefficient against MAP conditional growth is not a claim GT can be accused of having constructed.

(c) It is not a function of what the platform chose to serve. This is the deepest reason and it is the subject of the next subsection.

3.4 Circularity, and two distinct versions of it

A criterion that is itself a function of the admission decision makes the whole exercise circular: the study would be measuring how well the screener predicts its own consequences. There are two ways this can happen here, and they need different treatment.

Circularity of the first kind: the decision changes the treatment

If admitted children were treated differently *because* of the admission decision — placed in a faster track, given more staff attention, held to different expectations — then any criterion measured after admission is contaminated by the decision itself. A screener that predicts admission would then predict the criterion trivially.

The open-admission policy defuses this entirely, and this is an underrated second benefit of not rationing seats. If effectively every applicant is admitted, the screener score does not determine what happens next, because nothing about placement currently depends on it. The screener can be scored and sealed, and the programme proceeds as it would have anyway. This is a clean prospective design, and it is available only while the score is not in operational use.

Assumption to confirm. This holds only if the screener score is genuinely withheld from guides, from the platform, and from placement decisions during the validation window. If any part of the programme sees the score, circularity of the first kind returns. Confirm the score can be sealed operationally.

Circularity of the second kind: the platform routes the child

This one is live, and it is the reason the anchor exists.

In a personalised-mastery system, how fast a child clears objectives depends on which objectives they were given. If the platform’s routing responds to anything correlated with the screener signal, then mastery pace partly measures the routing, and the screener will appear to predict pace for a reason that is not about the child’s ability.

The distinction that matters is *what the routing responds to*:

- Routing that responds to the child’s *demonstrated performance on the platform* is benign. That is what a mastery system is supposed to do, and the resulting pace is a legitimate composite of ability and progress.
- Routing that responds to an *externally supplied label* — a prior CogAT score, a grade placement, a gifted flag, an enrolment tier — is not benign. If any such label correlates with the screener, it opens a path from screener to pace that bypasses the child entirely.

How much of the primary criterion’s validity rests on this assumption? It can be bounded. If routing R inflates the observed screener–pace correlation through a path $\rho_{SR} \cdot \rho_{RP}$, then to explain away an observed validity of 0.35 in full, that product must equal 0.35 — meaning both legs would have to be about 0.59. A routing variable correlating 0.59 with a sealed screener score, and 0.59 with pace net of everything else, is a strong and specific claim. It is not impossible; it is falsifiable. Appendix C tabulates the full sensitivity surface.

How large this effect can be: the simulated demonstration

Simulated data. The result below is a simulation of a mechanism, not a measurement of GT’s platform. Its value is in the comparison of magnitudes, not in the magnitudes themselves.

The companion analysis built a cohort, then varied nothing except how the platform routed children, and re-measured criterion validity under each routing regime. Mastery-pace validity drifted by +0.098 AUC. MAP conditional growth moved 0.000.

The routing-induced drift in mastery-pace validity is 2.9 times the entire effect this study is trying to detect (Section 5). A criterion that can move by three times the effect size for reasons that have nothing to do with the child is not a criterion to rely on alone. This is the argument for the external anchor, and it is quantitative rather than a matter of taste.

Claim boundary. The flatness of the MAP result is by construction, not a finding. MAP was built into the simulation as external to routing, so it could not move; the 0.000 is arithmetic, not evidence. What the simulation demonstrates is that the *mechanism* is large enough to matter and that a criterion outside the routing loop is immune *to the extent that it really is outside the loop*. Whether MAP genuinely is outside GT’s routing loop is an empirical question about GT’s platform, answered by the routing-input audit below and not by any simulation.

The check that would detect it. Contamination inflates the screener–pace correlation but cannot inflate the screener–MAP correlation, because MAP is outside the platform’s routing. So the diagnostic is the *gap* between the two:

$$H_0 : \rho(\text{screener}, \text{pace}) = \rho(\text{screener}, \text{MAP CGP}).$$

A materially larger correlation with pace than with MAP is the signature of routing contamination. This is a test of two dependent correlations sharing a variable, and it is powered on the cohort GT already has: detecting a gap between 0.45 and 0.30 needs 290 children, and the pooled 346 reaches power 0.87 (Appendix C).

Three supporting checks, in decreasing order of importance:

1. **Routing-input audit.** Ask the platform team, in writing, exactly which inputs the routing algorithm consumes. If no externally supplied ability label is among them, circularity of the second kind is largely closed by construction, and the statistical checks become confirmatory rather than load-bearing. This costs one conversation and is worth more than any amount of modelling.
2. **Exposure-adjusted validity.** Re-estimate the screener–pace relationship controlling for routing exposure (objectives served, mean difficulty served, minutes active, queue length). A large drop in the coefficient indicates the relationship runs through routing.
3. **Routing-predictability test.** Regress a routing summary on the sealed screener score. Some association is expected and benign; the question is whether it survives conditioning on the child’s own on-platform performance.

Assumption to confirm. All three supporting checks require routing/exposure telemetry at child level. Whether the platform retains this is currently unconfirmed. If it does not, the gap test against MAP remains available and is the one that matters most.

3.5 Measurement detail: do not regress on the raw percentile

Conditional growth is published as a percentile. Percentile ranks are approximately uniform by construction, and a uniform variable cannot correlate perfectly with a normally distributed predictor even under a perfect monotone relationship: the ceiling is $\sqrt{3/\pi} = 0.9772$. Using raw percentiles therefore discards about 4.5% of the explainable variance before the analysis begins. Apply a normal-quantile transform to the conditional growth percentile and analyse on that scale. Appendix A derives the constant.

4 Stage 1 — validity with no comparator

Stage 1 asks: *does the screener score predict mastery pace and MAP conditional growth?* It requires no CogAT data and can begin as soon as outcome data is joined.

4.1 Design

Score each enrolled child on the screener. Seal the score. At the end of the window, join to mastery pace over the same period and to MAP conditional growth autumn-to-spring. Estimate two correlations and their confidence intervals, and the gap between them (Section 3.4).

Because two primary criteria are tested, control the family-wise error rate; the sample-size tables report both the unadjusted and the Bonferroni-adjusted requirement.

4.2 What the existing cohorts can and cannot deliver

n	Smallest detectable r	95% CI if true $r = 0.35$	CI half-width	Power at $r = 0.35$	Power at $r = 0.25$
46	0.40	[0.07, 0.58]	0.26	0.67	0.39
100	0.28	[0.16, 0.51]	0.17	0.95	0.71
200	0.20	[0.22, 0.47]	0.12	> 0.99	0.95
300	0.16	[0.25, 0.45]	0.10	> 0.99	0.99
346	0.15	[0.25, 0.44]	0.09	> 0.99	> 0.99
500	0.13	[0.27, 0.42]	0.08	> 0.99	> 0.99
1000	0.09	[0.29, 0.40]	0.05	> 0.99	> 0.99

The 46-child campus cohort is a pilot, not a study. Its smallest detectable correlation is 0.40. If the true validity is 0.35, the interval it returns is [0.07, 0.58] — an interval that contains “almost useless” and “very good” simultaneously. Dummy-testing those children is still the right first move, because it de-risks administration, timing and scoring. It should not be presented as evidence of validity in either direction.

The pooled 346 children are a different matter: the smallest detectable correlation falls to 0.15 and the interval narrows to [0.25, 0.44], which is precise enough to act on. Reaching a half-width of ± 0.10 requires 298 children; ± 0.075 requires 528.

4.3 Plan for precision, not for significance

True r	n for 80% power	n , Bonferroni	n for CI ± 0.10
0.15	347	420	368
0.20	194	235	355
0.25	124	149	339
0.30	85	103	320
0.35	62	75	298
0.40	47	56	273
0.50	30	35	219

The two right-hand columns differ sharply, and the difference is the point. Detecting that validity is non-zero at $r = 0.35$ takes 62 children. Estimating it well enough to choose a cut score takes 298. Only the second is decision-relevant: nobody will set an admissions policy on the finding that validity is “not exactly zero.”

4.4 Clustering

Children are nested in campuses and guides, who share practices. Ignoring this inflates apparent precision. With an intraclass correlation of 0.05 and all 46 campus children under one roof, the design effect is 3.25 and the effective sample is 14.

Cluster size	ICC	Design effect	Effective n from 346
46	0.05	3.25	106
46	0.10	5.50	63
15	0.05	1.70	204
15	0.10	2.40	144
30	0.05	2.45	141
30	0.10	3.90	89

The practical implication favours GT Anywhere: a virtual cohort spread across many guides clusters far less than a single physical campus. Record a guide or cohort identifier for every child so this can be estimated rather than assumed.

4.5 Officer agreement — specified, pending data

Historical admission decisions exist but have not been released. The analysis is specified here so it can run the day they arrive.

Required record format, one row per historical applicant: applicant identifier (joinable to the screener); decision date; decision (admit / decline / waitlist / withdrawn); reviewing officer identifier; any rubric subscores recorded at the time; and the materials the officer saw. Cases reviewed by two or more officers must be identifiable as such, because they are the only rows that estimate inter-officer reliability.

Analysis. Quadratic-weighted κ between the screener’s implied decision and the officer’s actual decision; inter-officer κ on double-reviewed cases; and the association between officer decisions and the eventual criterion.

Claim boundary. Agreement with officers is not validity. If officers’ decisions have little predictive value, a screener that agrees with them perfectly also has little predictive value — high agreement would then be evidence *against* the screener, not for it. Officer agreement is a deployment-risk measure: it predicts how much disruption adopting the screener would cause. Inter-officer κ on double-reviewed cases is the more useful quantity, because it bounds how well *any* instrument could agree with a noisy human process.

What the simulation suggests this will look like

Simulated data. Simulated decisions, simulated officer. The pattern below is what the analysis is expected to *surface*; the numbers are not claims about GT’s actual admissions process.

A decision rule using the screener and MAP matched 86% of the simulated officer’s decisions, $\kappa = 0.43$. Read alone that looks mediocre. It is not the right comparison.

A rule fitted purely to *mimic* the officer, given her own stated inputs, ceilings at $\kappa = 0.79$. That is the upper bound on what any instrument could achieve, and the gap below 1.0 is the officer’s own inconsistency plus weight she places on things outside her stated rule. The screener-plus-MAP rule therefore reaches 54% of the achievable agreement, not 0.43 of a perfect standard. **Report agreement against the ceiling, never against 1.0** — doing otherwise holds the instrument to a standard the human process does not meet.

The disagreements are more informative than the agreement rate, because they are structured rather than random, and the two directions mean different things:

Files the rule would admit and she declined sit 0.92 SD lower on MAP reading. This is a *documented hard gate* that the statistical rule does not implement. It is not evidence that the rule is wrong; it is evidence that a declared policy constraint was left out of the model. The fix is to implement declared gates explicitly as gates rather than hoping a fitted rule recovers them — a hard threshold and a regression weight are different objects, and a regression will trade a low reading score against a high reasoning score in exactly the way the gate exists to forbid.

Files she admitted and the rule declined sit 0.29 SD higher on a construct-irrelevant impression once ability and MAP are partialled out. This is the finding that matters, and it is the one to handle carefully: it says some of the variance in her decisions is attributable to something that is not the construct the process is meant to select on. Whether that is legitimate expert judgement capturing something the measures miss, or construct-irrelevant variance in the sense of Messick (1995), cannot be settled by the agreement analysis. It is settled by asking whether that component predicts the criterion. If it does not, it is bias.

On the files where the two disagree, the rule’s picks reached the criterion 40% of the time against the officer’s 33%, with a pool base rate of 30%.

Claim boundary. Do not report that gap as a result. A difference of 7% between two proportions near a third needs about 742 disputed files *in each direction* to detect at 80% power. Disagreements are roughly one decision in seven, so that is on the order of 10600 historical decisions — far beyond what GT will have. The direction is suggestive and worth pre-registering as a hypothesis. It is not evidence, and at the sample sizes available it will not become evidence.

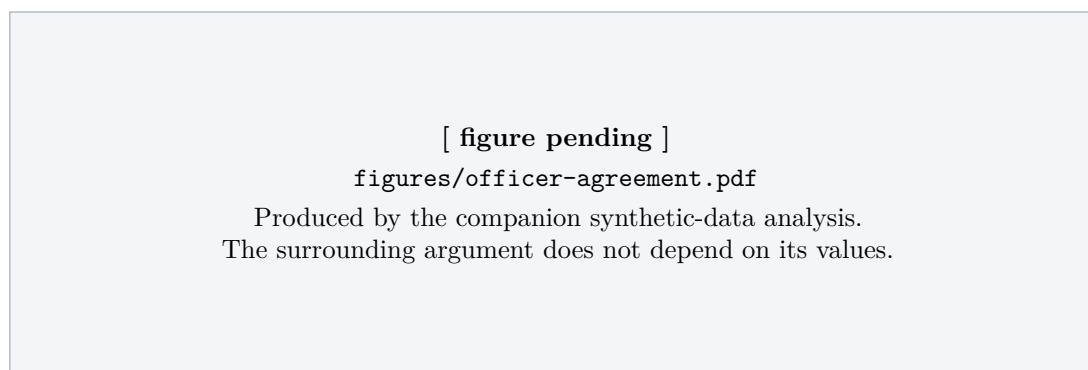


Figure 1: Agreement between screener-implied decisions and historical officer decisions, with the inter-officer agreement ceiling shown for reference. The ceiling matters more than the agreement itself: no instrument can agree with a human process more consistently than that process agrees with itself. Pending release of historical admission records (Section 4.5).

4.6 What Stage 1 licenses on its own

Claim boundary. After Stage 1, GT may say: “the screener predicts mastery pace and independently-measured academic growth, with validity estimated at r and a stated interval.” GT may *not* say that the screener is better than CogAT, because CogAT was not measured; nor that the screener caused anything; nor that children scoring highly benefited from the programme. A Stage 1 result is a statement about the screener, not a comparative or causal one.

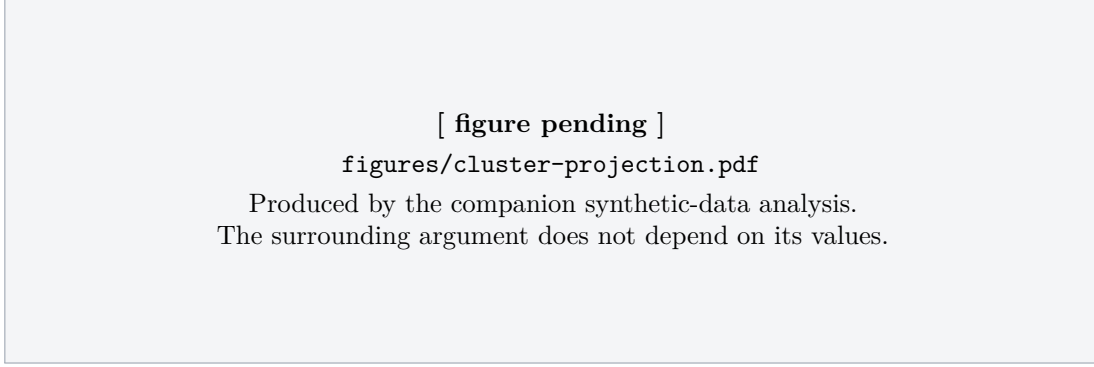


Figure 2: Low-dimensional projection of simulated applicant profiles, showing the structure the screener is intended to resolve and the degree to which candidate criterion groups separate in that space. Illustrative of the design’s logic on synthetic data; it carries no validity claim about real children.

5 Can the comparison be run at all?

Everything in Section 6 assumes the CogAT comparison is worth specifying. Companion simulation work has since measured the effect involved, and it is small enough that the question of whether GT can afford to detect it has to be settled before the design is settled. This section does that. It is placed here, before the Stage 2 design rather than in an appendix, because for a reader deciding whether to fund this work it is the most consequential section in the report.

Simulated data. The effect sizes below are simulated. Their *magnitude* is provisional; the *arithmetic* relating an effect size to a required sample is not, and would apply to any effect of similar size.

5.1 The constraint, stated plainly

The advantage to be detected is a difference of 0.034 in AUC ([+0.020, +0.049]) between the screener and a CogAT-like comparator. Subsampling the simulated cohort down to the size of one physical campus, 46 children, gives a 95% interval on a *single* AUC with a half-width of 0.169. The interval is 5.0 times the effect.

The analytic form corroborates the measurement. Hanley and McNeil’s variance for an AUC (Hanley & McNeil, 1982) gives a half-width of 0.179 at 46 children and a 30% base rate — within 0.01 of the subsampled value, which is reassuring about both. Table 1 extends it.

Table 1: Precision on a single AUC of 0.670 at a 30% base rate. The effect to be detected is 0.034. Derived in Appendix I.

n	95% half-width	vs. the effect	Full interval width
46	0.179	5.3×	0.357
100	0.120	3.5×	0.241
300	0.069	2.0×	0.139
346	0.065	1.9×	0.129
500	0.054	1.6×	0.107
1000	0.038	1.1×	0.076
2000	0.027	0.8×	0.054

5.2 A correction before despairing

That comparison is, however, the wrong one, and stating it that way overstates the difficulty. The half-width of 0.169 is the interval on *one* AUC estimated on its own. The quantity of interest is the *difference* between two AUCs measured on the same children, and paired differences are far more precise than the pair of marginal intervals suggests:

$$SE(A_1 - A_2) = \sqrt{SE_1^2 + SE_2^2 - 2r_{AUC} SE_1 SE_2},$$

where r_{AUC} is the correlation the two estimates inherit from the correlation between the two instruments (Hanley & McNeil, 1983). Two instruments that both measure reasoning are heavily correlated, and that correlation works in the study’s favour — it cancels out of the difference.

This matters by a factor of several, so it is worth being explicit: reading two overlapping confidence intervals and concluding “no difference” is a standard statistical error (Gelman & Stern, 2006), and the correct paired test is substantially more powerful. It is not, unfortunately, powerful enough.

Table 2: Sample required for the paired AUC comparison, by how strongly the two instruments correlate. Power 0.80, $\alpha = 0.05$ two-sided.

r_{AUC}	n for power 0.80	Power at 46	Power at 346
0.5	2590	0.07	0.18
0.7	1555	0.08	0.26
0.9	520	0.13	0.63

Even under the most favourable assumption — two instruments so alike that their AUC estimates correlate at 0.9 — the paired test needs 520 children, and reaches power 0.13 on one campus and 0.63 on every enrolled child GT currently has. If the instruments are merely moderately similar, the requirement is 1555 to 2590.

5.3 What the addressable sample actually is

46 is a cohort, not a ceiling, and the recommendation changes depending on which population the study draws from. Table 3 prices each route.

The raw head count is not the usable sample. Children are nested within campuses and guides, and children who share a learning environment have correlated outcomes. The usable sample is $n_{\text{eff}} = n/[1 + (m - 1)\rho_I]$, with m the cluster size and ρ_I the intraclass correlation (Killip et al., 2004). Pooling several campuses buys fewer independent observations than it appears to, and pooling one large campus buys very few.

Assumption to confirm. An intraclass correlation of 0.05 is assumed throughout Table 3; it is a common value in school-effects work but has not been estimated for this programme. It should be estimated from the first cohort’s data, because the routes differ in how much it costs them. The expansion route is the most exposed: its design effect is the largest in the table.

Each route costs something different:

One campus, one cycle (46). Maximum comparability — one site, one cohort, one version of the instrument, one set of guides. Minimum sample. *Effective n: 14*. This route cannot support any comparative claim.

Table 3: Addressable sample by route. “Increment” asks whether the route reaches 222 effective children, enough for the variance-increment test against the artefact floor; “AUC” asks whether it reaches 520, enough for the paired AUC comparison at $r_{\text{AUC}} = 0.9$. Enrolment figures are company-reported (E-101).

Route	Raw n	Design eff.	n_{eff}	Increment	AUC
One physical campus, one cycle	46	3.25	14	no	no
All physical campuses after planned expansion	246	3.45	71	no	no
GT Anywhere, one cycle	300	1.70	176	no	no
All enrolled children, one cycle	346	2.20	157	no	no
All enrolled, three consecutive cycles	1038	2.20	472	yes	no
Applicant pool at target scale	2000	2.20	909	yes	yes

All physical campuses after the planned expansion (≈ 246). GT intends to open several further campuses. Pooling them multiplies the head count but introduces the largest design effect in the table, because campuses are big clusters, and new campuses differ systematically from established ones in staff experience and cohort composition. *Effective n : 71* — fewer usable observations than the virtual cohort alone, from five times the children. This route is the trap in the table: it looks like the obvious way to grow the sample and it is close to the worst.

The virtual cohort, one cycle (300). The largest single clean population, with small clusters and therefore a modest design effect (*effective n : 176*), and — decisively — it is the unselective one, which is where the range-restriction advantage of Section 8 lives. It costs comparability against the physical campuses: delivery differs, the criterion is generated by the same platform but under different supervision, and a claim built here does not transfer to campus admissions without an invariance argument (Section 6).

Every enrolled child, one cycle (346). *Effective n : 157*. Requires establishing that the criterion means the same thing in both settings before pooling — a measurement-invariance question, not a convenience. This is the right Stage 1 population and it is still short of the Stage 2 requirement.

Three consecutive admission cycles ($\approx 1,038$). *Effective n : 472* — the first route that clears the variance-increment requirement. It costs time and it costs stability: the instrument must not change materially across the pooled cycles, the criterion definition must not change, and NWEA norm updates must be handled. Children who appear in more than one cycle contribute correlated observations and must be entered once, or modelled as repeated measures, not counted twice.

The applicant population at target scale ($\approx 2,000$). *Effective n : 909* — the only route that clears both requirements. It is also the only one where the binding cost is not enrolment: every child in a Stage 2 analysis must be *administered CogAT*, and at scale that is a procurement decision rather than a research one. GT should price it directly — the required n multiplied by the per-child cost of a CogAT administration is the actual budget for the head-to-head claim.

5.4 The estimand is a bigger lever than the sample

Before concluding that the comparison is unaffordable, it is worth noticing how much of the requirement is a consequence of how the question was posed rather than how big the effect is. Table 4 holds the simulated effect fixed and varies only the estimand.

Table 4: The same simulated effect, four ways of asking about it. Power 0.80, $\alpha = 0.05$. Derived in Appendix I.

Estimand	Effect	n required
Continuous criterion, ΔR^2 (screener added to comparator)	0.0495	140
Continuous criterion, ΔR^2 vs the artefact floor	0.0495	222
Criterion dichotomised at 30%, ΔR^2	0.0285	257
Paired AUC difference, $r_{\text{AUC}} = 0.9$	0.034	520
Paired AUC difference, $r_{\text{AUC}} = 0.7$	0.034	1555

Two decisions inside that table are free and together move the requirement by about a factor of 3.7.

Do not dichotomise the criterion. Mastery pace is continuous and was made continuous deliberately (Section 3). Splitting it into “excels / does not” at the 30% mark, which is what computing an AUC requires, multiplies the correlation by 0.759 and therefore discards about 42% of the explained variance (MacCallum et al., 2002). It costs a factor of 1.8 in sample. AUC is a good way to *communicate* an effect to a non-technical audience and a poor way to *estimate* one; report it, but do not test on it.

Ask the question GT actually needs answered. “Does the screener beat CogAT?” and “can the screener replace CogAT?” are different questions with different costs. The second is a non-inferiority question: it asks whether the screener is not worse by more than some margin GT is willing to accept.

Table 5: Non-inferiority at a 0.02 AUC margin against superiority, same simulated effect, power 0.80.

r_{AUC}	n for non-inferiority	n for superiority
0.5	1028	2590
0.7	617	1555
0.9	207	520

At 207 children the non-inferiority version is within reach of a pooled cohort today; the superiority version at 520 is not. The margin must be set in advance and justified as a difference GT would genuinely accept, not chosen after seeing the data — that is the one place a non-inferiority design can be abused, and a hostile reader will check it.

5.5 Verdict

At the programme’s current size, GT can establish that the screener predicts who thrives, and can plausibly establish that it is not worse than CogAT. GT cannot establish that it is better. The superiority claim needs 520–2590 children against an effective sample of 157; it becomes reachable across three pooled admission cycles for the variance increment, and only at applicant-population scale for the AUC comparison. This should shape what GT says publicly about the screener, starting now, rather than being discovered after the study reads out.

There is one further reason not to chase the superiority claim, which is that in the simulation it is not the interesting result. Adding the screener to the comparator raises explained variance by 0.0495; adding the comparator to the screener raises it by 0.0030 — below the floor that measurement error alone manufactures (Section 6.4). If that asymmetry survives contact with real data, the defensible claim is that CogAT contributes nothing the screener has not already captured. That is a replacement argument, it is worth more operationally than a marginal AUC advantage, and it is the cheaper one to establish.

6 Stage 2 — incremental validity over CogAT

Stage 2 begins when CogAT scores exist for a substantial subset of the validated cohort.

6.1 Why this is not an equating problem

There is a standing temptation to express screener scores “in CogAT units.” That would be equating, and it is not available here. Equating requires that two tests measure the same construct with comparable reliability and be interchangeable for the intended purpose; the resulting conversion must hold regardless of which population is examined. The GT screener and CogAT are adaptive versus fixed-form, differ in length and reliability, and are being designed to measure partly different things — which is the entire premise of building one.

What *is* available is concordance: a population-dependent statistical correspondence, honestly labelled as such. But concordance answers the wrong question. The question GT needs answered is not “what CogAT score is this equivalent to,” it is “does this tell us anything CogAT does not already tell us,” and that is an incremental-validity question. Appendix D sets out the distinction and its consequences.

6.2 The model

Hierarchical regression on the criterion Y :

$$\text{Step 1: } Y = \beta_0 + \beta_1 \text{CogAT} + \varepsilon \quad (1)$$

$$\text{Step 2: } Y = \beta_0 + \beta_1 \text{CogAT} + \beta_2 \text{Screener} + \varepsilon \quad (2)$$

The quantity of interest is ΔR^2 , tested by the F -test of the increment. The effect size to report is ΔR^2 itself alongside Cohen’s $f^2 = \Delta R^2 / (1 - R_{\text{full}}^2)$, plus the semipartial correlation. Report the increment with a confidence interval, not merely a p -value.

Add covariates for grade and age. If both screener components are entered, the step tests two predictors and the degrees of freedom change accordingly.

6.3 Sample size

Section 5 established the headline: the superiority version of this comparison is out of reach, and the variance-increment version is the affordable one. This subsection gives the general table so that the requirement can be recomputed when GT settles on an increment worth detecting.

ΔR^2	Screener alone ($q = 1$)			Both components ($q = 2$)		
	$R_{\text{CogAT}}^2 = 0.10$	0.25	0.40	0.10	0.25	0.40
0.01	701	583	466	861	716	572
0.02	348	289	230	427	355	283
0.03	230	191	152	283	235	187
0.05	136	112	89	167	138	110
0.08	83	68	54	102	84	66

Detecting a three-percentage-point increment over a CogAT baseline of $R^2 = 0.25$ requires 191 children; a five-point increment requires 112. The simulated increment for adding the screener to the comparator is 0.0495, which sits at the friendlier end of that table and needs 140 children — roughly what a pooled cohort supplies on a raw head count, though not after the design effect of Section 5 reduces 346 heads to 157 usable observations. And that figure holds only *if* the test is run against a null of zero. It should not be.

6.4 The artefact floor

CogAT is reliable but not perfectly so. Entering an imperfectly measured CogAT score into step 1 removes only part of the CogAT construct’s variance from the criterion. Whatever remains is available for step 2 to pick up — and a second measure of the *same* construct will pick it up.

Appendix E derives the size of this artefact in closed form. For two congeneric measures of one latent trait with reliabilities r_1 and r_2 , where the trait correlates ρ with the criterion and the second measure contributes *nothing new whatsoever*:

$$\Delta R_{\text{spurious}}^2 = \frac{\rho^2 r_2 (1 - r_1)^2}{1 - r_1 r_2}.$$

Comparator reliability r_1	$\rho = 0.31$	$\rho = 0.45$	$\rho = 0.63$
0.90	0.0046	0.0096	0.0188
0.94	0.0020	0.0043	0.0084
0.98	0.0003	0.0006	0.0012

Assumption to confirm. r_1 is the second of the two contestable assumptions flagged in Section 1.4. A working value of 0.94 is used here. It is not a verified figure: this project’s own evidence review attempted to confirm a published composite reliability from the primary technical documentation and could not, because the source PDFs were not text-extractable. The value was swept forward rather than asserted, and it is recorded that way in the evidence register. The table is therefore bracketed from 0.90 to 0.98 rather than quoted at a point, and the range it produces is not a rounding difference — the floor moves from 0.0003 to 0.0188, a factor of more than sixty across the corners of the table. *Estimate r_1 in GT’s own sample rather than inheriting it.*

At the working comparator reliability of 0.94 and a screener reliability of 0.90, a screener measuring *nothing* beyond CogAT still produces an expected increment of 0.0043 to 0.0084 of criterion variance. At 0.90 the floor rises to 0.0188. A study of 1376 children would detect the smaller figure at 80% power and 700 the larger, so at GT’s likely sample the naive test is not *guaranteed* to fire on the artefact alone — but the margin between a real increment and a manufactured one is thin, it depends on a reliability nobody has verified, and it moves in the wrong direction if the comparator is worse than assumed.

This is the practical form of a general result: statistically controlling for a construct by entering one fallible measure of it does not control for the construct.

6.5 The fix, and what it costs

Test against the floor, not against zero:

$$H_0 : \Delta R^2 \leq \Delta R_{\text{spurious}}^2(r_1, r_2, \rho) \quad \text{rather than} \quad H_0 : \Delta R^2 = 0.$$

Equivalently, require the lower bound of the confidence interval on ΔR^2 to exceed the floor. This is more demanding, and the cost is real:

True ΔR^2	n , null=0	n , null=floor	Inflation
0.03	191	380	2.0×
0.05	112	172	1.5×
0.08	68	89	1.3×
0.12	44	52	1.2×

Detecting a genuine five-point increment against the artefact floor takes 172 children rather than 112. A three-point increment takes 380, which is beyond the current cohort. **GT should therefore not design Stage 2 to detect small increments.** The defensible target is a large increment on a moderate sample, not a small increment on a sample that cannot distinguish it from measurement error.

Three supporting measures, in order of value:

1. **Report the floor alongside the estimate, always.** Whatever else is done, a ΔR^2 reported without its artefact floor is uninterpretable.
2. **Estimate r_1 locally** rather than relying on the publisher’s figure. If CogAT is administered once, use published reliability but treat it as an assumption and report sensitivity across the 0.90–0.98 range, as the table above does.
3. **Prefer a latent-variable model** where sample size allows. Modelling CogAT and the screener as indicators with fixed reliabilities removes the artefact by construction rather than by correction. This requires more data than GT will have at first, so it is the eventual form of the analysis, not the first one.

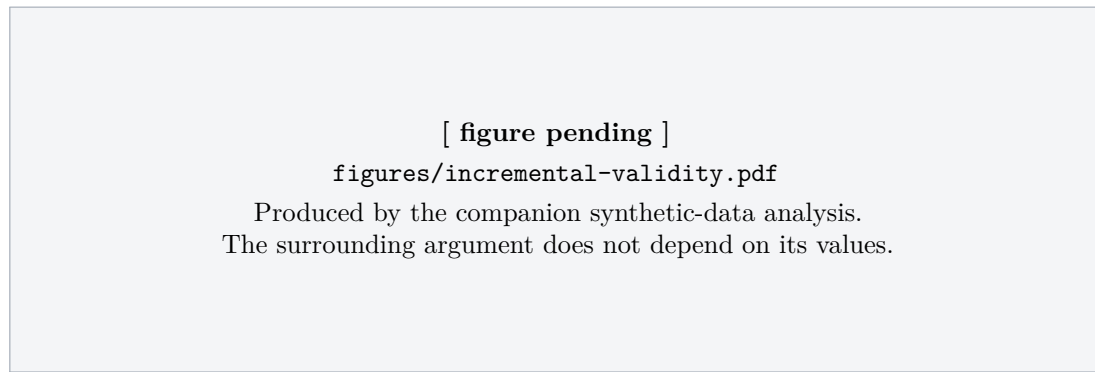


Figure 3: Variance in the criterion explained by CogAT alone and by CogAT plus the screener, with the increment shown against the artefact floor implied by CogAT’s own unreliability (Section 6.4). An increment that fails to clear the shaded floor is consistent with the screener adding nothing. Simulated data illustrating the comparison the real study will make.

6.6 What Stage 2 licenses

Claim boundary. After Stage 2, GT may say that the screener does or does not add predictive information beyond CogAT, by a stated amount, with a stated interval, against a stated artefact floor. GT may not say the screener is “better than CogAT” without naming the criterion and the population; validity is a property of a use, not of an instrument. A null result is a real and publishable finding: it would mean GT can adopt the cheaper, shorter instrument on equivalence grounds, which is a commercially useful result in its own right.

Claim boundary. At the sample sizes actually available, Stage 2 will very likely license the equivalence statement and not the superiority one (Section 5). This should be settled in advance rather than discovered at read-out, because the two conclusions want different pre-specified analyses, and switching to the equivalence framing after a superiority test fails is exactly the move a hostile reader will look for. Pre-register which question is being asked, with the non-inferiority margin fixed, before any CogAT score is linked to a criterion value.

7 Decision accuracy, not reliability

The screener’s output is a decision. The statistics that describe a decision are not the statistics that describe a score. Coefficient alpha is not among them: it measures internal consistency, is inflated by test length, and says nothing about whether the children placed above the cut are the right children.

7.1 Consistency and accuracy of classification

Two questions: would a second administration place the child on the same side of the cut (*decision consistency*), and does the observed classification match the one the child’s true score would produce (*decision accuracy*)?

Reliability	Consistency / accuracy at selection ratio			
	5%	10%	20%	50%
0.85	0.96 / 0.97	0.92 / 0.95	0.88 / 0.91	0.82 / 0.87
0.90	0.96 / 0.97	0.94 / 0.96	0.90 / 0.93	0.86 / 0.90
0.92	0.97 / 0.98	0.94 / 0.96	0.91 / 0.94	0.87 / 0.91
0.95	0.97 / 0.98	0.96 / 0.97	0.93 / 0.95	0.90 / 0.93

At a reliability of 0.90 and a top-10% cut, consistency is 0.94 and accuracy 0.96. Stated for a leader: **about 6% of children would land on the other side of the line if tested again next week.** That is not a failure of this instrument — it is what a reliability of 0.90 means, and it applies to CogAT too. It is an argument for reporting a confidence band with every score and for routing borderline cases to human review rather than to an automatic decision.

Note that consistency is highest at extreme cuts and lowest at the median (0.86 at a 50% cut), because few children sit near an extreme cut. A high consistency figure at a stringent cut is therefore not as impressive as it sounds and should not be quoted without the selection ratio.

7.2 Sensitivity, specificity and the arithmetic that binds

Once the criterion and the cut are fixed, sensitivity and specificity are not free parameters — they follow from the validity, the selection ratio and the base rate. Modelling the screener and criterion as bivariate normal (Appendix F):

Validity	Base rate	Sensitivity	Specificity	PPV	NPV
0.30	0.10	0.22	0.91	0.22	0.91
0.30	0.20	0.19	0.92	0.37	0.82
0.40	0.10	0.27	0.92	0.27	0.92
0.40	0.20	0.22	0.93	0.44	0.83
0.50	0.10	0.32	0.92	0.32	0.92
0.50	0.20	0.26	0.94	0.51	0.83
0.60	0.10	0.39	0.93	0.39	0.93
0.60	0.20	0.30	0.95	0.60	0.84

At a validity of 0.40, selecting the top 10%, with 10% of children meeting the “excels” criterion: positive predictive value is 0.27. **73% of the children the screener flags will not meet the criterion.** The screener is still doing real work — it multiplies the hit rate by 2.7 over selecting at random — but if the score is described to families as identifying children who will excel, that description is wrong for most of the children it identifies.

7.3 Base rate dominates

Holding the instrument fixed and moving only the prevalence of “excelling”:

Base rate	Se .80 / Sp .90	Se .90 / Sp .95	Se .95 / Sp .99
2%	0.14	0.27	0.66
5%	0.30	0.49	0.83
10%	0.47	0.67	0.91
20%	0.67	0.82	0.96
35%	0.81	0.91	0.98
50%	0.89	0.95	0.99

A very good instrument — sensitivity 0.90, specificity 0.95 — yields a PPV of 27% when only 2% of children excel, 49% at 5%, and 82% at 20%. The instrument did not change. Only the definition of “excels” changed.

The base rate is a choice GT makes when it defines the criterion, and it will do more to determine the screener’s apparent accuracy than any amount of psychometric work. Defining “excels” as the top 2% guarantees that most flagged children are false positives, no matter how good the test is. This should be decided deliberately, in advance, and with the consequence understood — not inherited from whatever threshold a report happens to use.

The same arithmetic run backwards: to reach a PPV of even 0.50 at a 5% base rate requires specificity of 0.953; at a 20% base rate, 0.775 suffices.

Base rate	Specificity for PPV=0.50	for PPV=0.75
2%	0.982	0.994
5%	0.953	0.984
10%	0.900	0.967
20%	0.775	0.925
35%	0.515	0.838
50%	0.100	0.700

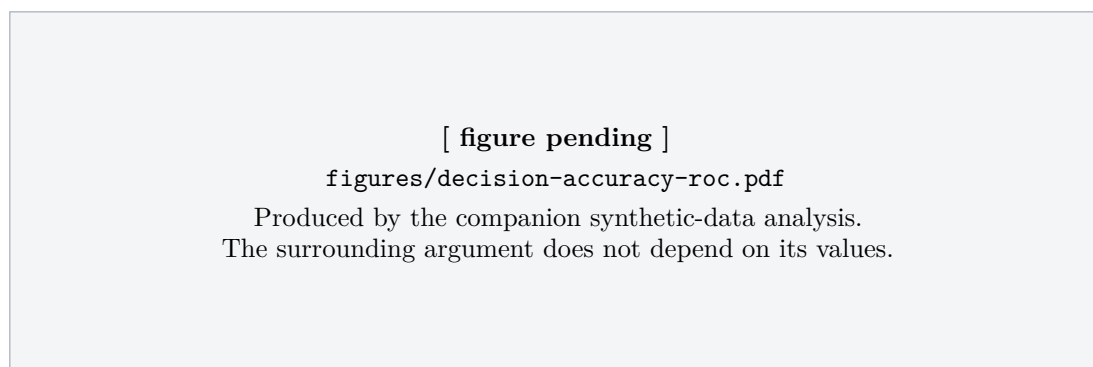


Figure 4: ROC curves for the screener and for CogAT against the ratified criterion, with the proposed operating cut marked on each. The area under the curve summarises ranking ability across all cuts; the marked point is what the programme would actually experience. Simulated data; the real comparison awaits Stage 2.

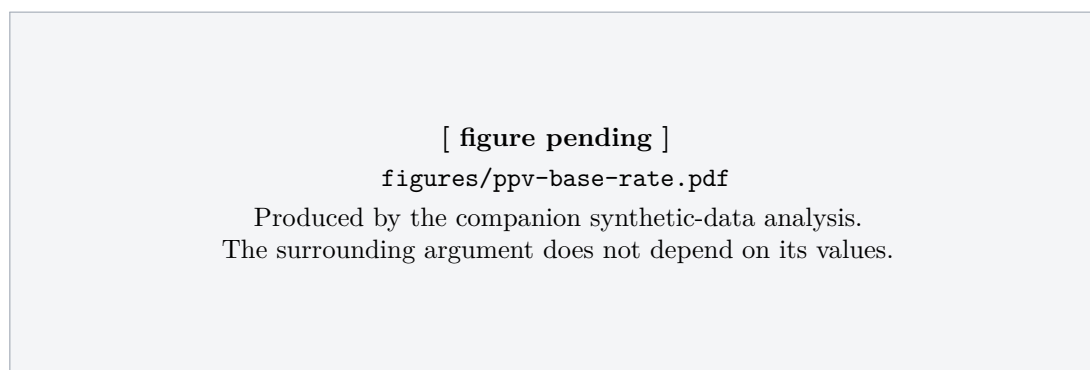


Figure 5: Positive predictive value as a function of the base rate of “excelling,” at fixed sensitivity and specificity. The curve is steep at low prevalence: the same instrument is trustworthy or untrustworthy depending on a definitional choice GT has not yet made (Section 7).

7.4 Both error types have owners

A false positive admits a child who then struggles — documented in GT’s own history, where children admitted on a non-CogAT “gifted” result were later described as struggling. A false negative denies a child who would have thrived. These costs fall on different people and are not interchangeable, and the ratio between them is a values judgement, not a statistical one. The design should report the full 2×2 table at several candidate cuts and let GT choose, rather than optimising a single figure of merit that hides the trade.

8 The range-restriction gift

8.1 The problem this programme does not have

Predictive-validity studies in selection are almost always crippled the same way. Outcomes are observed only for admitted students, and admission depended on the predictor. The observed correlation is computed in a truncated sample and is badly attenuated. Corrections exist — Thorndike’s Case II for direct restriction on the predictor, Case III for indirect restriction — but they require the unrestricted variance as an input, they assume linearity and homoscedasticity across a range that was never observed, and they are contested. Recent work found that standard corrections had systematically *overcorrected*, revising the operational validity of cognitive ability downward from roughly 0.51 to 0.31 — a revision large enough to overturn conclusions, produced entirely by the correction machinery.

GT currently does not ration seats. Effectively every applicant is admitted. Outcomes are observed across the full range of the predictor, and no correction is needed.

8.2 What it is worth

Selection	SD ratio	Observed r / n needed, by true ρ		
ratio	s/S	$\rho = 0.30$	$\rho = 0.40$	$\rho = 0.50$
5%	0.372	0.12 / 581	0.16 / 305	0.21 / 177
10%	0.411	0.13 / 475	0.18 / 250	0.23 / 145
20%	0.468	0.15 / 369	0.20 / 195	0.26 / 114
50%	0.603	0.19 / 224	0.25 / 119	0.33 / 71
100%	1.000	0.30 / 85	0.40 / 47	0.50 / 30

A true validity of 0.40 appears as 0.18 in a sample selected on the top 10% — an attenuation of 56%. Detecting it takes 250 children instead of 47: the open-admission policy is worth a 5.3 $\times$ reduction in required sample. This is what converts GT’s Stage 1 study from infeasible to comfortably feasible on the cohort that already exists.

Even mild selection is expensive: at a top-20% cut the multiplier is still 4.1 $\times$.

8.3 How to exploit it, deliberately

1. **Run the validity study now, while the window is open.** This is time-limited in a way that is easy to miss. It is the strongest methodological argument for prioritising Stage 1.
2. **Record the full predictor distribution**, including children who enrolled and left, and children who applied and did not enrol. Even under open admission, self-selection into applying is a form of restriction; documenting the applicant distribution against a reference population lets its size be estimated rather than assumed.
3. **Bank the unrestricted variance now.** The unrestricted standard deviation of the screener score, measured in an unselected cohort, is the input every future correction will need. It cannot be recovered later.
4. **Pre-register the estimate as uncorrected.** The headline validity should be the raw observed correlation, with no correction applied and none needed. That is a rare and defensible position and should be stated plainly.

8.4 If GT becomes selective

The design degrades in a predictable order. Direct restriction on the screener becomes Thorndike Case II, correctable if the unrestricted SD was banked. Selection on a composite that includes the screener becomes Case III, which requires the third-variable correlations as well. Selection on unrecorded judgement is the bad case: no correction is available, and the validity estimate becomes a lower bound of unknown tightness.

If selection begins, the mitigations in order of value are: keep scoring every applicant including those declined; keep the decision rule explicit and recorded, since an explicit rule is correctable and a judgement call is not; and if capacity is genuinely oversubscribed, randomise among borderline applicants — which would additionally create the only design in this document capable of supporting a causal claim.

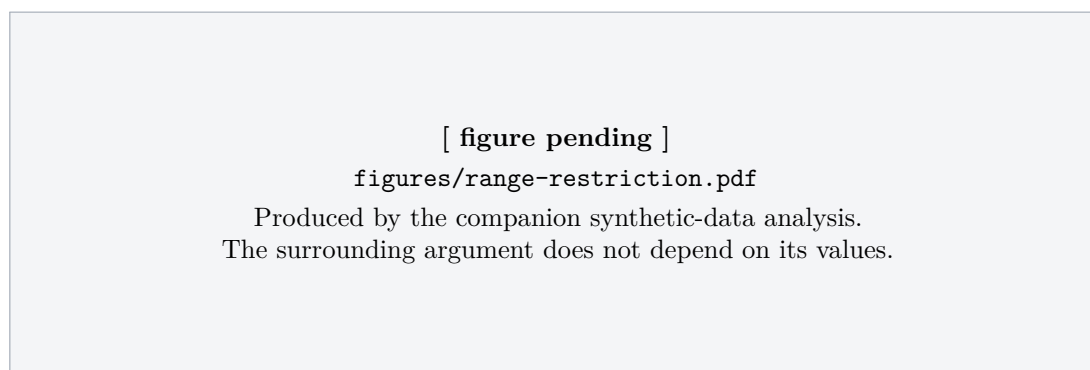


Figure 6: Attenuation of an observed validity coefficient as selectivity increases, with GT’s current position at the right-hand edge. The vertical distance between the curves and the horizontal true-validity line is the bias a conventional study would have to correct away and GT does not (Section 8).

9 What the two-phase structure buys

The instrument produces a standing ability estimate and, separately, a learning-rate estimate from a novel-task block. This section states what that second component can be claimed to do today, which is nothing, and what it would take to change that.

9.1 Current measured state

Recovery of an injected learning rate on the purpose-built bank, averaged over eight seeds, is 0.334 at the administered 30 trials, rising to 0.549 at 45 and 0.700 at 60. The contamination floor — the apparent learning rate recovered on a scrambled control arm, where no learning is possible — is non-zero and, critically, *does not fall with block length*. The smallest reference spread at which the readout can name a band is 0.075 even at 60 trials, against the only reference distribution with a stated provenance of 0.03; against that reference the readout is indeterminate for 100% of children at every block length measured.

Claim boundary. The learning-rate readout is not reportable at the administered block length. No claim in this report, and none in any GT-facing material, should assert that the instrument measures learning rate for an individual child. This is a measured result in GT’s own repository, not a cautious hedge.

9.2 What it would buy if it reached reportable precision

The honest case for a learning-rate component is that standing ability measures what a child has already acquired, and a selection instrument for an acceleration programme arguably wants to know how fast a child acquires *new* material. If those are separable, a learning-rate component would predict criterion variance that standing ability cannot reach, and would be especially valuable for children whose standing ability is depressed by prior opportunity rather than by capacity.

That is a hypothesis. The literature on dynamic assessment supports the construct’s coherence but not a specific increment, and the broader finding across process-signal research is that few studies run the decisive test at all: the literature is largely *silent* on incremental validity rather than supportive of it.

9.3 The right test: when does a subscore earn separate reporting?

This is exactly the subscore-value question. A subscore should be reported separately only when it carries information the total score does not, judged by proportional reduction in mean squared error. And there is a hard ceiling on how much it can add: the incremental validity of a subscore over the total is no greater than the reliability of the residual from linear prediction of the subscore by the total. Appendix H derives that ceiling for this instrument.

Trials	Recovery r	Implied reliability	Max overlap with standing	Ceiling on ΔR^2	Measured ΔR^2	n to detect
8	0.046	0.002	0.05	0.0019	—	—
15	0.179	0.032	0.19	0.0293	—	—
30	0.334	0.112	0.35	0.1025	0.007	836
45	0.549	0.301	0.58	0.2814	—	—
60	0.700	0.490	0.74	0.4648	0.015	387

Reading the row that describes the instrument as administered: at 30 trials the learning-rate estimate has an implied reliability of 0.112. It clears the subscore-value bar only if its true correlation with standing ability is below 0.35.

9.4 Two claims that must not be conflated

Simulated data. The increments below are simulated.

The companion analysis measured what the learning-rate readout contributes against the external MAP anchor: $\Delta R^2 = 0.007$ at the administered 30 trials, rising to 0.015 at 60. It also measured what lengthening the block does to the two sources of error separately, and the two behave completely differently.

	30 trials	60 trials	Change
Mean posterior SE (random error)	0.063	0.028	−56%
Contamination floor (systematic)	0.0097	0.0094	−3%

Doubling the block halves the random error and does essentially nothing to the contamination floor. That single fact separates two claims which are easy to run together and have entirely different evidentiary requirements.

Claim A — the learning rate is useful as a covariate. This asks only whether the readout *ranks* children in a way that carries criterion variance. Random error attenuates a ranking but does not bias it, so halving the posterior SE improves this claim directly, and the measured increment does rise from 0.007 to 0.015. A longer block buys Claim A.

Claim B — the instrument measures a child’s learning rate. This is an *absolute* claim about a quantity attached to a named child, and it is what anyone reading a score report will assume. It requires the contamination floor to be small relative to the reported quantity, because a floor is a systematic offset that does not average out and does not shrink with more trials. The floor falls by 3% across a doubling of block length. A longer block does not buy Claim B, and no feasible block length will.

Everything that follows about block length is therefore about Claim A only. Claim B is out of reach by a different mechanism than the one a longer block addresses, and it should be removed from the roadmap rather than deferred to a larger block.

9.5 Does the increment survive the artefact floor?

Even Claim A is marginal, and whether it survives depends on assumptions rather than on the measurement.

At 30 trials the measured increment of 0.007 sits above the 0.0043 artefact floor that comparator unreliability manufactures at the working reliability of 0.94 (Section 6.4) — but below the 0.0084 floor that applies if the criterion correlation is at the upper end of its plausible range. At 60 trials, 0.015 sits above the working floor. Detecting the 30-trial increment against a null of zero would take 836 children; the 60-trial increment would take 387. Neither is available (Section 5).

9.6 The assumption the whole component rests on

Assumption to confirm. This is the first of the two contestable assumptions from Section 1.4, and it is load-bearing for this entire section. The correlation between standing ability and learning rate is taken as 0.30. It has never been measured — not in this project, and not, so far as the evidence review found, anywhere. It is a working value chosen because it is plausible, not because it is supported.

The sensitivity is severe, and worse than it first appears.

$\rho(\text{standing, rate})$	30 trials		60 trials	
	Ceiling on ΔR^2	Subscore bar	Ceiling on ΔR^2	Subscore bar
0.00	0.1116	clears	0.4900	clears
0.15	0.1093	clears	0.4839	clears
0.30	0.1025	clears	0.4648	clears
0.50	0.0861	fails	0.4144	clears
0.70	0.0604	fails	0.3218	clears
0.90	0.0241	fails	0.1510	fails

At the administered block length the subscore-value bar is cleared only while the correlation stays below 0.35. The working assumption of 0.30 sits 0.05 below that threshold — and the component **already fails at 0.50**, not at the 0.90 that would represent near-redundancy. At 0.90 it fails at every block length in the table, including 60 trials, and the ceiling on its contribution falls to 0.0241.

A correlation of 0.5 between two cognitive constructs measured on the same children by the same instrument is not an extreme scenario; it is an ordinary one. **The two-phase argument is one modest empirical result away from being void, and that result has never been collected.** Measuring this correlation is the cheapest high-value experiment available to GT and should be prioritised above lengthening the block, because it determines whether lengthening the block is worth doing at all.

9.7 What follows

1. **Do not report a learning-rate score to any user.** Not as a number, not as a band, not as a hedge.
2. **Do keep administering the block during validation, and keep the raw trial data.** It costs test time but it is the only way to obtain the two unknowns that decide the component's fate — its true correlation with standing ability, and its true incremental validity — and those cannot be recovered retrospectively.
3. **Treat 60 trials as the minimum plausible reportable length,** and recognise that even there the contamination floor does not fall. The block length is necessary, not sufficient.
4. **Decide the block on test-time economics.** A 60-trial novel block inside a ≤ 45 -minute assessment competes directly with items measuring standing ability, which is measured well. The question is not whether learning rate is interesting; it is whether it is worth more than the standing-ability precision it displaces.

10 Minimum additional data

The owner wants this minimised. The good news is that most of what the design needs is believed to be collected already. The important caveat is that this belief is largely unverified.

Assumption to confirm. What the programme logs is currently the owner's expectation rather than a confirmed inventory. Every row below marked [A] is an assumption to confirm before the study is scheduled. If several prove wrong, the first-defensible-answer timeline changes materially.

#	Data element	Why the design needs it	Already ists?	ex-	Cost if not	Stage
1	Screener score per child, sealed	The predictor. Must be withheld from staff and platform during the window	Created by the study		Low	1
2	Mastery-pace log: objectives mastered with timestamps	The primary criterion	[A] logged	Believed	Fatal if absent	1
3	MAP RIT autumn + spring, and conditional growth percentile	The external anchor; the defence against “grading own homework”	Yes — administered 3×/yr		Low (export)	1
4	Stable child identifier joining screener, platform and MAP	Without it nothing can be linked	[A] Likely partial		Moderate — may need manual reconciliation	1
5	Enrolment start/end and active-days denominator	Pace is a rate; an uncorrected numerator confounds pace with time enrolled	[A] available	Believed	Low	1
6	Grade and date of birth	Conditioning, and age-appropriate norming	Near-certain		Negligible	1
7	Guide / campus / cohort identifier	Clustering; without it precision is overstated	[A] Likely		Low	1

— everything above is sufficient for a first defensible answer —

#	Data element	Why the design needs it	Already exists?	ex-	Cost if not	Stage
8	Routing and exposure telemetry: objectives served, difficulty, minutes, queue	The contamination checks in §3.4	[A] Uncertain; possibly partial		Moderate	1
9	Demographics: race, sex, EL status, free-lunch or equivalent	Differential prediction and fairness analysis; required before operational use	[A] Partially, via enrolment		Low	1
10	CogAT scores for the validated cohort	The entire comparator. Does not exist. Needs 222 effective children for the increment, 520+ for superiority (§5)	No — must be gathered		High, and it scales with n : licensing, proctoring, staff time	2
11	A repeat CogAT administration on a subsample ($n \approx 100$)	Estimates the comparator's reliability locally instead of inheriting the unverified 0.94. That figure sets the artefact floor every Stage 2 conclusion is measured against (§6.4)	No		Moderate — one extra session for a subsample	2
12	Historical admission decisions with officer identifiers	Officer-agreement analysis (§4.5)	Exists, not yet released		Low once released	2
13	Double-reviewed historical cases, flagged	The only rows that estimate inter-officer reliability, which bounds the whole officer analysis	[A] Unknown		Low	2
— everything above is sufficient for a strong answer —						
14	Second screener administration on a subsample ($n \approx 100$)	Empirical decision consistency instead of the modelled figures in §7	No		Moderate — one retest session	2
15	Guide ratings collected blind to the screener score	A fallback criterion, and convergent evidence for pace	No		Moderate — staff time	2
16	Multi-year mastery pace and MAP for the same children	Distinguishes durable prediction from a first-year effect	Accrues time	with	Time only	3

Top three by necessity. Rows 1–3. The screener score, the mastery-pace log, and MAP growth. With those three plus an identifier that joins them, Stage 1 runs. Everything else improves the answer or extends it; only these make an answer possible at all.

Row 11 is small and disproportionately valuable. It is the only row that converts a contestable assumption into a measurement. Without it, every Stage 2 result is reported against a floor inherited from a figure this project could not verify, and a hostile reader gets to choose which end of the range to argue from. One repeat administration on a hundred children closes that.

The single largest risk in this table is row 4. A validity study is a join. If children cannot be matched across the screener, the platform and MAP with high confidence, sample size is lost precisely where it is scarcest, and match failure is unlikely to be random — it will correlate with enrolment churn, which correlates with the criterion. Confirm the join key before anything else.

11 Open questions

These are recorded rather than answered. Inventing answers to them would be the main way this document could mislead.

On the criterion

1. Is “excels” a threshold or a rank? This sets the base rate, which Section 7 shows dominates everything else about decision quality.
2. Over what window is mastery pace computed, and how are partial enrolments, absences and holidays handled in the denominator?
3. Are objectives comparable in size across subjects and grades? If a mathematics objective takes twice as long as a reading objective, pace measures subject mix as much as speed.
4. Which MAP subject is the anchor — mathematics, reading, or a composite? This must be pre-declared, or the analysis has an undisclosed multiplicity problem.
5. Is there a floor or ceiling in the pace measure — children who master nothing, or who exhaust available content?

On routing and contamination

6. **What inputs does the platform’s routing algorithm actually consume?** The single highest-value question in this list. If no external ability label is among them, circularity of the second kind is largely closed by construction.
7. Is routing telemetry retained at child level, and for how long?
8. Can the screener score be genuinely sealed from guides and from the platform for the duration of the validation window?

On CogAT

9. Which form and level, administered when, and by whom? Form and level affect reliability and hence the artefact floor in Section 6.4.
10. Will CogAT be administered to the whole validated cohort or a subsample? If a subsample, how selected — because selecting on anything related to the criterion reintroduces the range restriction the design otherwise escapes. The number of administrations is also the budget line that decides whether Stage 2 happens at all (§5).
11. **What is CogAT’s composite reliability in a sample like GT’s?** This report uses 0.94 as a working value because an attempt to verify a published figure from the primary technical documentation failed. It should be estimated locally rather than inherited. It moves the artefact floor by a factor of more than sixty across the corners of the sensitivity table (§6.4).
12. **A discrepancy to resolve:** the evidence register records a GT commitment to supply “every student’s CogAT and MAP screeners,” while the owner states CogAT data does not yet exist. Both may be true — a commitment to gather is not a holding of data — but the register entry should be corrected so it is not later read as a claim that the data was already available.

On outcome data

13. Which of the rows marked [A] in Section 10 actually exist, at what granularity, and with what history?
14. Is there historical mastery-pace data for children already enrolled? If so, a retrospective Stage 1 could run immediately — at the cost that those children’s screener scores would have to be collected now and related to past outcomes, which reverses the temporal order and weakens the design.

On admissions history

15. When will historical decisions be released, and in what format (§4.5)?
16. Were any applicants declined? If genuinely none were, the officer-agreement analysis has no variance to explain and should be dropped rather than run.
17. Is the programme likely to become selective during the validation window? If so, Section 8 becomes urgent rather than advisory.

On the instrument

18. What is the true correlation between learning rate and standing ability? Unmeasured, and it determines whether the learning-rate block can ever earn separate reporting (§9). The value used in this report’s tables, 0.30, is an illustrative assumption and nothing more.
19. What is the screener’s own test–retest reliability? The 0.90 used throughout is a design target, not a measurement, and every artefact-floor figure depends on it.

On scale and design parameters

20. **Does GT need a superiority claim or a replacement claim?** This is a business decision, not a methodological one, and it is the largest single driver of cost: 520–2590 children against 207 (§5). It should be answered before anything else in this list.
21. **Which population will Stage 2 draw from,** and has the comparability cost of that route been accepted? Table 3 prices each; the expansion route in particular looks better than it is.
22. **What is the intraclass correlation of the criterion within campuses and within guides?** Assumed at 0.05 throughout. It converts head count into usable sample and no route’s verdict in Table 3 is robust to getting it badly wrong. Estimable from the first cohort at no additional collection cost.
23. **How strongly do the screener and CogAT correlate?** It sets r_{AUC} , which moves the required sample for the paired comparison by a factor of five (§5). Estimable from the first children who take both.
24. If GT wants a non-inferiority claim, **what AUC margin would GT genuinely accept** in exchange for whatever the screener offers over CogAT? This must be fixed before data are seen, and justified in terms of what GT gains — cost, time, breadth, accessibility — not chosen to make the result come out.

Part III — Technical appendices

A Criterion scale: the cost of regressing on a percentile

Conditional growth is published as a percentile rank. Suppose the underlying growth construct G is normally distributed and the screener relates to it linearly. The published criterion is $P = \Phi(G)$, which is uniform on $(0, 1)$. Even a perfect monotone relationship between G and P is not a perfect linear one, so regressing on P loses information.

The size of the loss is the correlation between a standard normal variate and its own probability integral transform. Integrating by parts with $u = \Phi(z)$ and $dv = z\varphi(z) dz$, so $v = -\varphi(z)$:

$$\mathbb{E}[Z \Phi(Z)] = \left[-\Phi(z)\varphi(z) \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \varphi(z)^2 dz = 0 + \frac{1}{2\pi} \int e^{-z^2} dz = \frac{1}{2\sqrt{\pi}}.$$

Since $\mathbb{E}[Z] = 0$ this is also $\text{Cov}(Z, \Phi(Z))$, and $\text{Var}(\Phi(Z)) = 1/12$ because $\Phi(Z)$ is uniform. Hence

$$\text{corr}(Z, \Phi(Z)) = \frac{1/(2\sqrt{\pi})}{1 \cdot \sqrt{1/12}} = \frac{\sqrt{12}}{2\sqrt{\pi}} = \sqrt{\frac{3}{\pi}} = 0.9772.$$

The attainable R^2 is therefore multiplied by $3/\pi = 0.9549$: about 4.5% of explainable variance is discarded by using the percentile directly. Applying Φ^{-1} to the conditional growth percentile before analysis recovers it. The constant is verified numerically against quadrature in `selftest.py`.

B Stage 1: precision and power for a correlation

For a bivariate normal sample, $z = \text{artanh}(r)$ is approximately normal with standard error $1/\sqrt{n-3}$. A two-sided test of $H_0: \rho = 0$ at level α with power $1 - \beta$ against a true ρ requires

$$n = \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\text{artanh } \rho} \right)^2 + 3,$$

and the confidence interval is obtained by transforming $\text{artanh}(r) \pm z_{1-\alpha/2}/\sqrt{n-3}$ back through $\tanh$. The routine reproduces Cohen's tabled value of $n = 85$ for $\rho = 0.30$ exactly; at $\rho = 0.50$ it returns 30 against Cohen's 28, the normal approximation being mildly conservative at small n , which errs toward larger samples and is therefore safe for planning.

Why the report emphasises interval width over power. A significance test answers whether validity is distinguishable from zero. No admissions policy turns on that. The decision-relevant quantity is the width of the interval, because the cut score and the expected decision accuracy both depend on the magnitude. This is why Section 4 reports the sample needed for a ± 0.10 interval (298 children at $r = 0.35$) alongside the much smaller sample needed for significance (62).

Multiplicity. Two primary criteria are tested. The tables report the Bonferroni-adjusted requirement ($\alpha = 0.025$) as well as the unadjusted one. Bonferroni is conservative here because the two criteria are positively correlated; it is used because the correlation between them is unknown, and a conservative adjustment is easier to defend than an estimated one.

Clustering. With clusters of size m and intraclass correlation ICC, the variance of an estimate is inflated by the design effect $1 + (m - 1)\text{ICC}$, and the effective sample is n divided by that factor.

C Criterion contamination: bounds and the detection test

C.1 How large a confounder would have to be

Let S be the sealed screener score, P mastery pace, and R a routing variable. If R is the only path from S to P other than the child’s ability, the contribution of that path to the observed correlation is $\rho_{SR}\rho_{RP}$. To account for an observed r_{SP} in full, the product of the two legs must equal r_{SP} ; the symmetric case sets both legs to $\sqrt{r_{SP}}$.

Observed r_{SP}	Both legs equal	ρ_{RP} if $\rho_{SR}=0.30$	ρ_{RP} if $\rho_{SR}=0.50$
0.20	0.45	0.67	0.40
0.30	0.55	1.00	0.60
0.35	0.59	> 1 (impossible)	0.70
0.40	0.63	> 1 (impossible)	0.80
0.50	0.71	> 1 (impossible)	1.00

At an observed validity of 0.35, a confounder would need both legs at 0.59, or — if routing correlates only 0.30 with the screener — a routing-to-pace association above 1, which is impossible. The reading is that *weak* routing dependence on the screener cannot explain a moderate validity; only strong dependence can. That is a specific, checkable claim rather than an unfalsifiable worry.

This is the logic of sensitivity analysis for unmeasured confounding: rather than asserting no confounding, state how strong confounding would have to be, and let the reader judge whether that is plausible.

C.2 The detection test

Contamination inflates $\text{corr}(S, P)$ but not $\text{corr}(S, M)$ where M is MAP conditional growth, because MAP is outside the platform’s routing. The diagnostic is therefore the difference between two correlations that share the variable S , tested by the Meng–Rosenthal–Rubin statistic:

$$Z = (z_1 - z_2) \sqrt{\frac{n - 3}{2(1 - r_{PM})h}}, \quad h = \frac{1 - f\bar{r}^2}{1 - \bar{r}^2}, \quad f = \min\left(\frac{1 - r_{PM}}{2(1 - \bar{r}^2)}, 1\right),$$

with $z_i = \text{artanh}(r_i)$ and $\bar{r}^2 = (r_1^2 + r_2^2)/2$. Required samples:

$r(\text{screener, pace})$	$r(\text{screener, MAP})$	n for 80% power, by $r(\text{pace, MAP})$		
		0.30	0.50	0.70
0.45	0.30	398	290	179
0.40	0.30	919	668	408
0.40	0.25	421	306	187
0.35	0.20	441	319	195

Note the direction of the dependence on r_{PM} : the more strongly pace and MAP agree, the *easier* the discrepancy is to detect, because the comparison becomes a within-child contrast. Detecting a gap between 0.45 and 0.30 at $r_{PM} = 0.50$ requires 290 children; the pooled cohort of 346 reaches power 0.87. GT Anywhere alone (300 children) reaches 0.81, so the check does not depend on pooling the two cohorts — which matters, because pooling a virtual and a physical cohort introduces exactly the clustering discussed in Appendix B.

C.3 Interpreting the outcome

- **No detectable gap.** Pace and MAP agree about the screener. Contamination is bounded below the detectable difference, and the primary criterion can be used with its full validity estimate.
- **Pace materially exceeds MAP.** Consistent with routing contamination, and also with pace being a more proximal and less noisy measure of what the screener predicts. These are not distinguishable by this test alone; the exposure-adjusted analysis in Section 3.4 is what separates them. Until it does, the MAP figure should be treated as the headline.
- **MAP exceeds pace.** Not contamination. Most likely the pace measure is noisy, or its denominator is mis-specified.

Claim boundary. This test bounds contamination; it does not eliminate it. A routing mechanism that happened to affect pace and MAP equally would be invisible to it. That is implausible given MAP’s externality but is not excluded, and it is the reason the routing-input audit in Section 3.4 is listed ahead of the statistical checks.

D Equating, linking and concordance

The measurement literature distinguishes several kinds of relationship between scores on two tests, in decreasing order of strength.

Equating produces interchangeable scores. It requires that the two tests measure the same construct, at comparable reliability, with the same intended use, and that the conversion be invariant across subpopulations. Only then may a score on one be substituted for a score on the other without qualification.

Calibration and projection are weaker forms, applicable when tests measure the same construct at different reliabilities or difficulties.

Concordance is a purely statistical correspondence between scores on tests measuring *different* constructs. It is population-dependent: the concordance table derived in one group need not hold in another. It supports statements of the form “in this population, children at this screener score tended to obtain that CogAT score,” and nothing stronger.

Why this matters for GT. A concordance table between the screener and CogAT would be straightforward to produce once both are administered, and it would be actively harmful if labelled an equating. It would invite the inference that the screener can substitute for CogAT in contexts where CogAT is required — district gifted identification, for instance — which the evidence would not support and which would expose GT to a defensible complaint.

More importantly, concordance answers a question GT does not need answered. Knowing that a screener score of X corresponds to a CogAT SAS of Y tells GT nothing about whether the screener should replace CogAT. That decision rests on which instrument better predicts the

criterion, and on whether the screener adds information CogAT lacks. Both are incremental-validity questions, which is why Section 6 is framed as it is.

The invariance question is separate and still required. Before operational use, the screener must be checked for differential item functioning and for measurement invariance across the groups it will be used on, and for differential prediction — whether a given score forecasts the same criterion value in every group. A test can be free of item bias and still predict differently across groups; these are distinct properties requiring distinct evidence.

E The increment manufactured by unreliability

E.1 Setup

Let T be a latent ability, standardised. Let the criterion Y be standardised with $\text{corr}(T, Y) = \rho$. Let two instruments be congeneric indicators of T with reliabilities r_1 and r_2 :

$$X_1 = \sqrt{r_1} T + \sqrt{1 - r_1} e_1, \quad X_2 = \sqrt{r_2} T + \sqrt{1 - r_2} e_2,$$

with e_1, e_2 independent of each other, of T , and of Y given T . By construction X_2 carries *no information about Y beyond T* : it is a second thermometer for the same temperature. Write

$$a = \text{corr}(X_1, Y) = \sqrt{r_1} \rho, \quad b = \text{corr}(X_2, Y) = \sqrt{r_2} \rho, \quad c = \text{corr}(X_1, X_2) = \sqrt{r_1 r_2}.$$

E.2 Derivation

R^2 from X_1 alone is $a^2 = r_1 \rho^2$. For two predictors,

$$R_{\text{full}}^2 = \frac{a^2 + b^2 - 2abc}{1 - c^2}.$$

Substituting, $a^2 + b^2 = \rho^2(r_1 + r_2)$ and $2abc = 2\rho^2 \sqrt{r_1 r_2} \sqrt{r_1 r_2} = 2\rho^2 r_1 r_2$, so the numerator is $\rho^2(r_1 + r_2 - 2r_1 r_2)$ and

$$\begin{aligned} \Delta R^2 &= \frac{\rho^2(r_1 + r_2 - 2r_1 r_2)}{1 - r_1 r_2} - r_1 \rho^2 = \rho^2 \frac{r_1 + r_2 - 2r_1 r_2 - r_1(1 - r_1 r_2)}{1 - r_1 r_2} \\ &= \rho^2 \frac{r_2 - 2r_1 r_2 + r_1^2 r_2}{1 - r_1 r_2} = \boxed{\rho^2 \frac{r_2(1 - r_1)^2}{1 - r_1 r_2}} \end{aligned} \tag{3}$$

The closed form is checked in `selftest.py` against R^2 computed directly from the 3×3 correlation matrix, at several parameter combinations, to machine precision.

E.3 Reading it

The expression behaves as it should. It vanishes when $r_1 = 1$: a perfectly measured incumbent leaves nothing for a redundant challenger to pick up. It grows with ρ^2 , so the artefact is largest exactly when the construct matters most. It grows with r_2 , so a *more* reliable redundant challenger produces a *larger* spurious increment — which is the counterintuitive part, and the

reason a well-built screener is more likely to produce a misleading positive than a poorly built one.

Substituting GT’s figures — comparator reliability 0.90–0.98, a screener reliability target of 0.90, and a criterion correlation between the Sackett-revised operational value 0.31 and the observed meta-analytic value 0.63 — gives the table in Section 6.4: an artefact between 0.0003 and 0.0084, with 0.0043 as the central case.

E.4 Consequence for the test

At the central case, the conventional test against $\Delta R^2 = 0$ reaches 80% power to declare the *artefact* significant at $n = 1376$; at the pessimistic case, at $n = 700$. Since Stage 2 will plausibly run at a few hundred children, a naive analysis is at material risk of reporting a significant increment that is entirely an artefact of CogAT’s measurement error. Testing against the floor instead raises the sample needed for a real five-point increment from 112 to 172.

Assumption to confirm. The floor depends on r_2 , the screener’s own reliability, which is currently a design target of 0.90 rather than a measurement. It also assumes the two instruments are congeneric indicators of a single latent trait, which is the worst case for GT — it is the assumption under which the screener adds nothing. If the screener genuinely measures a partly distinct construct, the true floor is lower and the test is conservative.

F Decision accuracy and the base-rate arithmetic

F.1 The 2×2 table is implied, not chosen

Sensitivity and specificity are often treated as properties of an instrument. For a continuous score cut at a threshold they are not: they follow from the validity, the selection ratio and the base rate. Model the screener X and the criterion Y as standard bivariate normal with correlation ρ . Select children with $X > z_x$ where $z_x = \Phi^{-1}(1 - \text{SR})$; define “excels” as $Y > z_y$ where $z_y = \Phi^{-1}(1 - \text{BR})$. Then

$$\begin{aligned} \text{TP} &= \Pr(X > z_x, Y > z_y), & \text{FP} &= \text{SR} - \text{TP}, & \text{FN} &= \text{BR} - \text{TP}, \\ \text{TN} &= 1 - \text{SR} - \text{FN}, & \text{Se} &= \frac{\text{TP}}{\text{BR}}, & \text{Sp} &= \frac{\text{TN}}{1 - \text{BR}}, & \text{PPV} &= \frac{\text{TP}}{\text{SR}}. \end{aligned}$$

The joint probability is a bivariate normal orthant, computed here by one-dimensional quadrature and validated against Sheppard’s theorem, against Plackett’s identity, and against the published Taylor–Russell tables.

A structural consequence worth noting: sensitivity is bounded above by SR/BR . If GT selects the top 10% but 20% of children meet the criterion, sensitivity cannot exceed 0.50 however good the instrument is, because there are not enough places. The low sensitivity figures in Section 7 are largely this, not instrument failure.

F.2 The odds form, as a cross-check

The same arithmetic in likelihood-ratio form makes the base-rate dependence transparent:

$$\underbrace{\frac{\text{PPV}}{1 - \text{PPV}}}_{\text{posterior odds}} = \underbrace{\frac{\text{BR}}{1 - \text{BR}}}_{\text{prior odds}} \times \underbrace{\frac{\text{Se}}{1 - \text{Sp}}}_{\text{likelihood ratio}}.$$

The instrument contributes only the likelihood ratio. At $\text{Se} = 0.90$ and $\text{Sp} = 0.95$ the likelihood ratio is 18, which is strong. At a base rate of 2% the prior odds are 1:49, so the posterior odds are 18:49 — a PPV of 0.27. **A likelihood ratio of 18 is an excellent diagnostic and still leaves most flagged children below the bar**, because the prior was so unfavourable. No improvement in the instrument fixes this; only redefining the criterion, or accepting that the score surfaces candidates rather than identifies successes, does.

F.3 Decision consistency and accuracy

Model observed score $X = T + E$ with $\text{Var}(X) = 1$ and $\text{Var}(T) = r_{xx}$. Two parallel administrations correlate r_{xx} , so consistency at a cut c is $\Pr(X_1 > c, X_2 > c) + \Pr(X_1 < c, X_2 < c)$ under a bivariate normal with correlation r_{xx} . Accuracy compares the observed classification to the true-score classification at the same raw cut, using $\text{corr}(X, T) = \sqrt{r_{xx}}$ and the cut $c/\sqrt{r_{xx}}$ on the standardised true score.

At a median cut this has the exact closed form $\frac{1}{2} + \arcsin(r_{xx})/\pi$, which `selftest.py` verifies to nine decimal places. These are model-based figures assuming normality and classical error; the operational estimate for a real test would use the Livingston–Lewis procedure on the actual score distribution. The modelled figures are adequate for design, and the report labels them as modelled.

G Range restriction

G.1 Thorndike Case II

Under direct selection on the predictor with unrestricted SD S and restricted SD s , write $U = S/s$. The observed correlation in the restricted group relates to the unrestricted one by

$$r_{\text{restricted}} = \frac{\rho/U}{\sqrt{1 - \rho^2 + \rho^2/U^2}}, \quad \rho = \frac{r_{\text{restricted}} U}{\sqrt{1 - r_{\text{restricted}}^2 + r_{\text{restricted}}^2 U^2}}.$$

For top-fraction selection with ratio p and cut $c = \Phi^{-1}(1 - p)$, the truncated variance follows from the standard normal moments. With $\lambda = \varphi(c)/p$ the inverse Mills ratio,

$$\text{Var}(X \mid X > c) = 1 + \lambda c - \lambda^2.$$

At $p = 0.10$: $c = 1.2816$, $\lambda = 1.7550$, and the SD ratio is $s/S = 0.411$. The closed form is checked against numerical quadrature of the truncated density in `selftest.py`, and the attenuation and correction functions are checked to invert one another.

G.2 What GT avoids

A true validity of 0.40 observed under top-10% selection appears as 0.18, an attenuation of 56%. The sample needed to detect it at 80% power rises from 47 to 250 — a factor of 5.3. At top-20% selection the factor is still 4.1.

The deeper benefit is that no correction is applied at all. Corrections require the unrestricted SD as an input, assume linearity and homoscedasticity across a range that was by definition never observed, and have been shown to systematically overcorrect: the operational validity of cognitive ability was revised downward from roughly 0.51 to 0.31 once the overcorrection was identified. **An uncorrected estimate from an unrestricted sample is not merely more efficient — it is a different and stronger class of evidence, because it does not depend on the correction machinery that produced that error.**

G.3 Indirect restriction, if it appears

If selection operates on a composite that includes the screener rather than on the screener alone, Case II no longer applies and Case III is required, which additionally needs the correlations between the screener, the composite and the criterion. Case III corrections are substantially less stable. The mitigation is procedural rather than statistical: keep the selection rule explicit and recorded. An explicit rule is correctable; an unrecorded judgement is not.

H Block length and the subscore-value bar

H.1 From recovery to reliability

The repository reports *recovery* r : the correlation between the estimated learning rate and the injected true value in simulation. If the estimate is the truth plus independent error, then $\text{corr}(\hat{\lambda}, \lambda) = \sqrt{\text{reliability}}$, so the implied reliability of the readout is r^2 . At the administered 30 trials, recovery of 0.334 implies a reliability of 0.112; at 60 trials, 0.700 implies 0.490.

H.2 The subscore-value bar

A subscore earns separate reporting only when it predicts its own true score better than the total does — the proportional-reduction-in-mean-squared-error criterion. Writing r_s for the subscore reliability, r_x for the total's, and ρ for the correlation between their true scores, the condition is $r_s > \rho^2 r_x$, so the subscore clears the bar only when

$$\rho < \sqrt{r_s/r_x}.$$

At 30 trials this ceiling is 0.35: the learning-rate block earns separate reporting only if learning rate and standing ability are correlated below 0.35 in truth. At 60 trials the ceiling relaxes to 0.74.

H.3 The hard ceiling on incremental validity

Separately, there is an upper bound on how much a subscore can add to prediction of an *external* criterion: the incremental validity is no greater than the reliability of the residual from linear prediction of the subscore by the total. With S and X standardised, $\text{corr}(S, X) = c = \rho\sqrt{r_s r_x}$, the residual is $e = S - cX$ with $\text{Var}(e) = 1 - c^2$. Its true part is $T_e = T_s - cT_x$, and since $\text{Var}(T_s) = r_s$, $\text{Var}(T_x) = r_x$ and $\text{Cov}(T_s, T_x) = c$,

$$\text{Var}(T_e) = r_s - 2c^2 + c^2 r_x = r_s [1 - \rho^2 r_x (2 - r_x)],$$

$$\Delta R_{\max}^2 = \text{rel}(e) = \frac{r_s[1 - \rho^2 r_x(2 - r_x)]}{1 - \rho^2 r_s r_x}.$$

At $\rho = 0$ this reduces to r_s : an orthogonal subscore can add at most its own reliability, which at 30 trials is 0.112. At $\rho = 0.30$ the ceiling falls to 0.1025.

H.4 Sensitivity to the unmeasured correlation

Both bars — the subscore-value bar $\rho < \sqrt{r_s/r_x}$ and the incremental ceiling above — are governed by ρ , the true correlation between learning rate and standing ability, which has never been measured. The sensitivity is tabulated in Section 9. Two features of it deserve emphasis here, because both are easy to miss from the main text.

First, the failure point at the administered block length is $\rho \approx 0.35$, and the working assumption of 0.30 sits only 0.05 below it. The component does not survive to 0.90 and then fail; it fails at 0.50, which is an unremarkable correlation between two cognitive constructs measured on the same children by the same instrument on the same day.

Second, the two bars fail in different ways and the difference matters for how a negative result should be reported. Exceeding the subscore-value bar means the subscore should not be reported *at all*, because the total score predicts it better than it predicts itself. Falling below the incremental ceiling means only that the subscore's contribution is bounded and small. The first is a prohibition; the second is a magnitude. At 0.90 the administered block fails the first, which is the stronger and more damaging outcome.

H.5 Attenuation, and the decisive comparison

The ceiling is an upper bound. The *expected* increment is governed by attenuation: a predictor with reliability r_s realises only r_s of a true ΔR^2 . A genuinely real increment of 0.05 is realised as $0.112 \times 0.05 = 0.0056$ at 30 trials.

The companion simulation measured the realised increment directly, against the external MAP anchor, and obtained 0.007 at 30 trials and 0.015 at 60. The agreement with the attenuation prediction at 30 trials is close, which is mild corroboration that the attenuation model is describing the right mechanism.

Whether the measured increment clears the artefact floor depends on an assumption, not on the measurement. At the working comparator reliability of 0.94, the floor is 0.0043 and the 30-trial increment of 0.007 sits above it. If the criterion correlation is at the upper end of its plausible range the floor rises to 0.0084 and the same increment sits below it. If the comparator's reliability is 0.90 rather than 0.94, the floor reaches 0.0188 and the increment is not close.

The honest summary is that the learning-rate component's contribution at the administered block length is *the same order of magnitude as the artefact a second fallible measure of the same construct produces for free*. It cannot be separated from that artefact without first pinning down the comparator's reliability in GT's own sample — which is a measurement GT can make, and should, before running this comparison at all.

At 60 trials the measured increment rises to 0.015, above the working floor, and detectable at 387 children rather than 836. Sixty trials is therefore the minimum plausible length for the covariate claim — necessary, and still not sufficient for an absolute claim, because the contamination floor falls only 3% across that doubling while the random error falls 56% (Section 9.4).

Assumption to confirm. The correlation between learning rate and standing ability, $\rho = 0.30$ in these tables, is an illustrative value. It has not been measured and it drives both the subscore-value ceiling and the incremental ceiling. Measuring it is the main reason to keep administering the block during validation even though its output is not reportable.

I ROC precision, paired comparison, and the cost of the frame

This appendix supports Section 5. It derives the precision of an AUC estimate, the precision of the *difference* between two AUCs measured on the same children, and the translation between the AUC frame and the variance frame used everywhere else in this report.

Precision of a single AUC

For an AUC of A estimated on n_1 positive and n_0 negative cases, Hanley and McNeil (1982) give

$$\text{Var}(A) = \frac{A(1-A) + (n_1-1)(Q_1-A^2) + (n_0-1)(Q_2-A^2)}{n_1 n_0},$$

$$Q_1 = \frac{A}{2-A}, \quad Q_2 = \frac{2A^2}{1+A},$$

where Q_1 and Q_2 are the probabilities that two positives both outrank a negative, and that a positive outranks two negatives. The expression is exact under an exponential model for the score distributions and is used here as an approximation.

Two checks are run against this in `scripts/selftest.py`. At $A = 0.5$ it must reduce exactly to the null variance of the Mann–Whitney statistic, $\text{Var} = (N+1)/(12n_1n_0)$, which it does to machine precision; and at $A = 1$ it must vanish, which it does. At $n = 46$ with a 30% base rate it returns a half-width of 0.179, against 0.169 measured by subsampling the simulated cohort. The agreement is close enough to treat either as the working figure.

Why the paired difference is far more precise

The relevant quantity is not the width of either marginal interval but the precision of $A_1 - A_2$ when both are computed on the *same* children:

$$\text{Var}(A_1 - A_2) = \text{Var}(A_1) + \text{Var}(A_2) - 2r_{\text{AUC}}\sqrt{\text{Var}(A_1)\text{Var}(A_2)},$$

with r_{AUC} the correlation between the two AUC statistics, itself induced by the correlation between the two instruments' scores within the positive and negative groups (Hanley & McNeil, 1983). Because both instruments measure overlapping reasoning constructs, r_{AUC} is large and the shared sampling error cancels.

This is the reason Section 5 does not simply divide the single-AUC half-width by the effect. Comparing two marginal intervals and concluding “no difference” where they overlap is a well-documented error (Gelman & Stern, 2006). The correct paired test is several times more efficient — and still, at $r_{\text{AUC}} = 0.9$, requires 520 children for 80% power against an effect of 0.034.

r_{AUC} is not known for this pair of instruments and is treated as a sensitivity parameter throughout, spanning 0.5 to 0.9. It can be estimated directly once any sample has both scores, and doing so early is worthwhile: it moves the required sample by a factor of five.

Non-inferiority

A superiority test asks whether the lower bound on $A_1 - A_2$ exceeds zero. A non-inferiority test asks whether it exceeds $-\delta$ for a pre-specified margin δ . With a true difference Δ the required sample follows from

$$\frac{\Delta + \delta}{\text{SE}(A_1 - A_2)} \geq z_{1-\alpha} + z_{1-\beta},$$

one-sided at $\alpha = 0.025$. At $\delta = 0.02$ this gives 207 children at $r_{\text{AUC}} = 0.9$ against 520 for superiority. The margin must be fixed before the data are seen and justified as a difference GT would genuinely tolerate in exchange for whatever the screener offers over CogAT — cost, time, breadth, or accessibility. A margin chosen afterwards to make the result come out is the standard abuse of this design and is easy for a reviewer to detect from the analysis timestamp.

Translating between AUC and R^2

The rest of this report works in correlations and variance increments; the companion simulation reports AUCs. The bridge, for a continuous predictor X and a latent criterion Y that are bivariate normal with correlation ρ , with the binary outcome defined as Y exceeding the $1 - p$ quantile c :

$$\text{AUC} = \int_{-\infty}^{\infty} f_+(x) F_-(x) dx, \quad f_+(x) = \frac{\phi(x) \bar{\Phi}\left(\frac{c - \rho x}{\sqrt{1 - \rho^2}}\right)}{p},$$

and f_- the same with Φ in place of $\bar{\Phi}$ and $1 - p$ in the denominator. This is evaluated by quadrature in `scripts/statlib.py` and inverted by bisection.

It is validated against an independent reduction. Writing $U = X_i - X_j$, the event $\{X_i > X_j, Y_i > c, Y_j \leq c\}$ is an orthant probability of the trivariate normal $(U/\sqrt{2}, Y_i, -Y_j)$ with correlations $(\rho/\sqrt{2}, \rho/\sqrt{2}, 0)$; conditioning on the first coordinate reduces it to a one-dimensional integral over the bivariate orthant routine already validated against Sheppard's theorem and Plackett's identity. The two paths agree to 10^{-6} .

Applying it to the simulated results: an AUC of 0.670 at a 30% base rate corresponds to $\rho = 0.36$, and 0.636 to $\rho = 0.29$. The first of those is worth noting — the validity of 0.35 assumed throughout this report as a working value, chosen before the simulation was run, lands within 0.01 of what the simulation produced.

What dichotomising the criterion costs

Computing an AUC requires a binary criterion. Mastery pace is continuous, so using AUC means dichotomising it. For a bivariate normal pair split at base rate p , the correlation with the dichotomised criterion is

$$r_{\text{pb}} = \rho \frac{\phi(c)}{\sqrt{p(1-p)}},$$

whose multiplier is at most $\sqrt{2/\pi} = 0.798$ at a median split and 0.759 at $p = 30\%$. Squaring, about 42% of the explained variance is discarded — information the study paid to collect and then threw away at the analysis step (MacCallum et al., 2002). In sample terms the penalty is a factor of 1.8, and the further step to a paired AUC comparison costs a factor of 3.7 against the continuous variance increment.

The recommendation that follows is narrow and worth stating precisely: compute AUC, report AUC, and put AUC in the slide deck, because it is the statistic a non-technical audience can interpret. Do not *test* on it, and do not *size the study* on it.

The asymmetry between the two increments

Finally, the two directions of the incremental question are not the same question and do not have the same answer. In the simulation:

$$R_{\text{comparator}}^2 = 0.084, \quad R_{\text{screener}}^2 = 0.130, \quad R_{\text{both}}^2 = 0.133.$$

Adding the screener to the comparator gains 0.0495; adding the comparator to the screener gains 0.0030, which falls below the artefact floor of Appendix E. A result of that shape does not license “the screener is better.” It licenses “the comparator adds nothing measurable once the screener is in the model,” which is a claim about redundancy rather than superiority, and is the claim Section 5 recommends GT pursue.

J Reproducibility

J.1 How the numbers were produced

Every numeric value in this report is generated by `scripts/compute.py` and written into `generated/` as $\text{\LaTeX}$ macros and table bodies, which the document then includes. No number is typed into the prose by hand. To regenerate:

```
cd scripts && python3 selftest.py && python3 compute.py
```

`scripts/statlib.py` implements the statistical routines using only the Python standard library — no NumPy, no SciPy — so the results reproduce on a bare Python 3.8+ interpreter. The routines include the regularised incomplete beta function, the central and noncentral F distributions, bivariate normal orthant probabilities, and the power and sample-size functions built on them.

J.2 Validation

`scripts/selftest.py` checks those routines against external reference values and internal consistency identities, and exits non-zero on any failure. It is a prerequisite of the build. The external checks include:

- central F quantiles against published tables at three (df_1, df_2, α) combinations;
- two G*Power worked examples for the R^2 -increment test, reproduced exactly ($n = 77$ for $f^2 = 0.15$ with $q = 3$; $n = 395$ for $f^2 = 0.02$ with $q = 1$);
- Cohen’s tabled sample size for a correlation ($n = 85$ at $\rho = 0.30$);
- Taylor–Russell tabled success ratios at two parameter combinations;
- Sheppard’s theorem for the bivariate normal at the median cut, across five correlations;
- Plackett’s identity for off-centre orthant probabilities;
- the Hanley–McNeil AUC variance reduced to the Mann–Whitney null variance $(N + 1)/(12n_1n_0)$ at $A = 0.5$, exactly, at three sample splits;
- the AUC/correlation bridge against an independent trivariate-orthant reduction of the same probability, agreeing to 10^{-6} (Appendix I).

Internal checks confirm that the range-restriction correction inverts the attenuation exactly, that the closed-form spurious increment matches R^2 computed from the correlation matrix, that the $\sqrt{3/\pi}$ constant matches quadrature, that a zero-validity predictor selects exactly at the base rate, and that a zero-reliability test agrees with itself only at chance.

J.3 Building the document

`make` regenerates the numbers and builds `main.pdf`. If no local L^AT_EX installation is present, `make docker` builds the same PDF in a TeX Live container. `make check` runs the self-test and a consistency check that every macro referenced in the sources is defined in `generated/numbers.tex`. That check catches three things the build either misses or reports late: a stale `generated/` directory, a macro that is computed but no longer referenced anywhere (usually a number left behind by a rewritten passage), and a passage calling a macro that no longer exists.

J.4 Figures

The six figures are produced by a companion synthetic-data analysis and are included via `\figslot`, which falls back to a labelled placeholder when the PDF is absent. The document therefore builds with or without them, and no argument in the text depends on a figure’s values.

J.5 Limits of these computations

Three caveats a reviewer should hold onto. The power calculations assume bivariate normality and homoscedasticity; mastery pace is a rate and is likely right-skewed, so the analysis should either transform it or use a model appropriate to counts, and the sample sizes here should be treated as lower bounds. The noncentrality convention is $\lambda = f^2n$, matching G*Power and verified against its worked examples; other conventions give slightly different answers and the difference is not negligible at small n . And every figure depending on the screener’s reliability inherits the status of that number, which is a design target rather than a measurement.

A fourth caveat is more consequential than the other three and is stated in the main text as well. The two parameters that most strongly drive the conclusions — the correlation between standing ability and learning rate, and the comparator’s reliability — are unmeasured working values, not estimates. They are carried through every table that depends on them as explicit sensitivity ranges rather than points, and the computations are structured so that revising either is a one-line change in `scripts/compute.py` followed by a rebuild. A reader who disputes either value can therefore recompute the report’s numbers under their own assumption rather than arguing about the conclusion.

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Repository sources

Claims about the instrument's current state are grounded in the project's own records rather than external literature: `docs/product/STAGE2_BANK_RECOVERY_MEASUREMENT.md` (recovery ladder, contamination floor, indeterminate rates); `docs/product/STAGE2_QUESTION_DESIGN.md` (block design and gates); `docs/research/ASSUMPTIONS_AND_EVIDENCE.md` entries E-095, E-100, E-101, E-200 (recovery, validation-data commitment, cohort sizes, estimator defect); `docs/governance/DECISION_LOG.md` entries D-030 and D-200 (learning-rate reportability); `docs/research/IN_HOUSE_COGNITIVE_TEST_RESEARCH.md` and `docs/research/EVIDENCE_DOSSIER.md` (CogAT and MAP psychometric properties); `docs/interviews/CRYSTAL_MARTEL_INTERVIEW_NOTES.md` (cohort sizes, seat rationing, validation-data availability).